

UNIVERSITY OF BOTSWANA

DEPARTMENT OF MATHEMATICS

Examination: Semester I

Year: 2020/21

COURSE CODE: MAT 122 DURATION: 1 Hour 30 Minutes DATE: DEC 2020

TITLE OF PAPER: INTRODUCTORY MATHEMATICS II

SUBJECT: MATHEMATICS TITLE OF EXAMINATION: BSc I

INSTRUCTIONS:

- Answer all questions in SECTION A.
- In SECTION B answer one (1) out of two (2) questions.

No. OF PAGES: 4

DO NOT OPEN THIS PAPER UNTIL YOU HAVE BEEN TOLD TO
DO SO BY THE INVIGILATOR.

SECTION A

Answer All Questions

QUESTION 1

- (a) Find the standard form of the equation of the parabola with vertex $(1, 0)$ and focus $(2, 0)$. [5]

- (b) Find the limit: [5]

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+5} - \sqrt{5}}{x}.$$

QUESTION 2

Find the derivative of the following functions:

- (a)

$$y = \frac{2}{\sqrt{x}} + x \sin x,$$

[5]

- (b)

$$g(x) = e^{x^2} \tan(x) + \ln(x^2 + 1).$$

[5]

QUESTION 3

- (a) Find $\frac{dy}{dx}$ implicitly if

$$x^2 y^3 - xy = 10.$$

[5]

- (b) Find the derivative of

$$f(x) = 7x - 3$$

[5]

by the limit process.

QUESTION 4

(a) Evaluate the integral

$$\int \frac{1+x}{\sqrt{x}} dx.$$

[4]

(b) Show that

$$\int_0^{\frac{\pi}{3}} \frac{\sin x}{(1 + \cos x)^2} dx = \frac{1}{6}.$$

[6]

QUESTION 5

(a) Find the equation of the tangent line to the graph of

$$f(x) = \sqrt{x^2 + 3}$$

at $x = -1$.

[8]

(b) Evaluate:

[2]

$$\lim_{x \rightarrow 0} \sin \left(\frac{x^3 + \pi}{2} \right).$$

QUESTION 6

Find the coordinates of the center, vertices, and the foci for the hyperbola with equation

$$4x^2 - 9y^2 - 32x - 18y + 19 = 0.$$

[10]

SECTION B

Answer ONE Question OUT of TWO
Each Question carries 20 Marks

Question 7

Given

$$f(x) = 3x^4 - 4x^3.$$

- (a) Find the critical numbers of f . [4]
- (b) Find the open intervals on which f is increasing or decreasing. [6]
- (c) Identify where the extrema of f occur. [3]
- (d) Find the intervals on which f is concave up and where it is concave down. [7]

Question 8

- (a) Use integration by parts to evaluate

$$\int x^2 \sin x \, dx.$$

[10]

- (b) Show how that

$$\int_1^9 \frac{1}{\sqrt{x}(1 + \sqrt{x})^2} \, dx = \frac{1}{2}.$$

[10]

MAT 122 Exam Solutions

①

①(a) $(y-k)^2 = 4p(x-h) \checkmark$

$h=1, k=0 \text{ and } p=2-1=1 \checkmark$

$(y-0)^2 = 4(1)(x-1) \Rightarrow y = 4(x-1) \checkmark \checkmark \quad [5]$

(b) $\lim_{x \rightarrow 0} \frac{\sqrt{x+5} - \sqrt{5}}{x} \cdot \frac{\sqrt{x+5} + \sqrt{5}}{\sqrt{x+5} + \sqrt{5}} \checkmark \checkmark$

$= \lim_{x \rightarrow 0} \frac{x+5-5}{x(\sqrt{x+5} + \sqrt{5})} \checkmark$

$= \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{x+5} + \sqrt{5})}$

$= \lim_{x \rightarrow 0} \frac{1}{\sqrt{x+5} + \sqrt{5}} \checkmark$

$= \frac{1}{\sqrt{5} + \sqrt{5}} = \frac{1}{2\sqrt{5}} \checkmark \quad [5]$

(2)

$$2(a) \quad y = 2x^{-1/2} + x \sin x$$

$$\begin{aligned} \frac{dy}{dx} &= 2(-1/2)x^{-3/2} + \sin x + x \cos x \\ &= -\frac{1}{x^{3/2}} + \sin x + x \cos x \quad [5] \end{aligned}$$

$$\begin{aligned} (b) \quad g'(x) &= e^{x^2} \sec^2 x + 2x e^{x^2} \tan x + \frac{1}{x^2+1} \cdot 2x \\ &= e^{x^2} \sec^2 x + 2x e^{x^2} \tan x + \frac{2x}{x^2+1} \quad [5] \end{aligned}$$

$$(3)(a) \quad 2xy^3 + x^2 \cdot 3y^2 \frac{dy}{dx} - (y + x \frac{dy}{dx}) = 0$$

$$3x^2y^2 \frac{dy}{dx} - x \frac{dy}{dx} = y - 2xy^3$$

$$\frac{dy}{dx} = \frac{y - 2xy^3}{3x^2y^2 - x} \quad [5]$$

$$(b) \quad f'(x) = \lim_{h \rightarrow 0} \frac{7(x+h) - 3 - (7x - 3)}{h} = \lim_{h \rightarrow 0} \frac{7x + 7h - 3 - 7x + 3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{7h}{h} = 7 \quad [5]$$

③

$$\textcircled{4} (a) \int \frac{1+x}{\sqrt{x}} dx = \int (x^{-1/2} + x^{1/2}) dx \checkmark \checkmark$$

$$= \frac{x^{1/2}}{1/2} + \frac{x^{3/2}}{3/2} + C \checkmark \checkmark$$

$$= 2\sqrt{x} + \frac{2}{3}x^{3/2} + C \checkmark \quad [5]$$

$$(b) u = 1 + \cos x \Rightarrow du = -\sin x dx \Rightarrow -du = \sin x dx \checkmark \checkmark$$

$$\int \frac{-du}{u^2} = - \left[\frac{u^{-1}}{-1} \right] = \frac{1}{u} + C \checkmark$$

$$= \frac{1}{1 + \cos x} \Big|_0^{\pi/3}$$

$$= \frac{1}{1 + 1/2} - \frac{1}{1 + 1}$$

$$= \frac{1}{3/2} - \frac{1}{2}$$

$$= \frac{2}{3} - \frac{1}{2} = \frac{4-3}{6} = \frac{1}{6} \checkmark \checkmark$$

[5]

(4)

$$\textcircled{c} (a) f' = \frac{1}{2}(x^2+3)^{-1/2} \cdot 2x \checkmark \checkmark$$
$$= \frac{x}{\sqrt{x^2+3}}$$

$$f'(-1) = \frac{-1}{\sqrt{1+3}} = -\frac{1}{2} \checkmark \checkmark$$

$$f(-1) = \sqrt{1+3} = 2 \checkmark \checkmark$$

[8]

$$y-2 = -\frac{1}{2}(x+1) \checkmark \checkmark$$

$$(b) \lim_{x \rightarrow 0} \sin\left(\frac{x^3+\pi}{2}\right) = \sin\left(\frac{\pi}{2}\right) = 1 \checkmark [2]$$

(5)

$$(6) \quad 4(x^2 - 8x) - 9(y^2 + 2y) = -19 \quad \checkmark$$

$$4(x^2 - 8x + 4^2) - 9(y^2 + 2y + 1^2) = -19 + 4(16) - 9 \quad \checkmark$$

$$4(\cancel{x^2} - x - 4)^2 - 9(y + 1)^2 = 36 \quad \checkmark$$

$$\frac{(x-4)^2}{9} - \frac{(y+1)^2}{4} = 1 \quad \checkmark$$

$$a=3, b=2 \Rightarrow c = \sqrt{9+4} = \sqrt{13} \quad \checkmark$$

$$h=4, k=-1 \Rightarrow \text{center } (4, -1) \quad \checkmark$$

$$\text{Vertices } (7, -1), (1, -1) \quad \checkmark \checkmark$$

$$\text{Foci } (1 \pm \sqrt{13}, -1) \quad \checkmark \checkmark$$

1) $f'(x) = 12x^3 - 12x^2 \checkmark \checkmark$

(a) $f'(x) = 0 \Rightarrow 12x^3 - 12x^2 = 0$

$\Rightarrow 12x^2(x-1) = 0$

$\Rightarrow x=0, x=1$ are critical numbers [4]

(b)

Interval	$-\infty < x < 0$	$0 < x < 1$	$1 < x < \infty$
Test value	$x = -1$	$x = \frac{1}{2}$	$x = 2$
Sign of $f'(x)$	$- \checkmark$	$- \checkmark$	$+ \checkmark$
Conclusion	decreasing $\checkmark$	decreasing $\checkmark$	increasing $\checkmark$

[6]

(c)

$x=1$ is a minimum value. $\checkmark \checkmark \checkmark$ [3]

(d) $f''(x) = 36x^2 - 24x = 0 \checkmark \checkmark$

$\Rightarrow 6x(6x-4) = 0$

$\Rightarrow x=0, x = \frac{2}{3} \checkmark \checkmark$

[7]

$-\infty < x < 0 \Rightarrow f''(-1) > 0 \Rightarrow \text{concave upward} \checkmark$

$0 < x < \frac{2}{3} \Rightarrow f''(\frac{1}{2}) < 0 \Rightarrow \text{concave downward} \checkmark^{9-12}$

$\frac{2}{3} < x < \infty \Rightarrow f''(1) > 0 \Rightarrow \text{concave upward} \checkmark$

(7)

$$\cup (a) \quad u = x^2 \Rightarrow du = 2x dx \quad \checkmark$$

$$dv = \sin x dx \Rightarrow v = \int \sin x dx = -\cos x \quad \checkmark$$

$$\int x^2 \sin x dx = x^2 \cos x - \int \cos x \cdot 2x dx \quad \checkmark \checkmark$$

↓

$$u = 2x \Rightarrow du = 2 dx \quad \checkmark \checkmark$$

$$dv = \cos x dx \Rightarrow v = \sin x$$

$$\int x^2 \sin x dx = x^2 \cos x - \left[-2x \sin x + \int \sin x (2 dx) \right] \quad \checkmark \checkmark$$

$$= x^2 \cos x + 2x \sin x + 2 \cos x + C \quad \checkmark \checkmark \quad [10]$$

$$(b) \quad u = 1 + \sqrt{x} = 1 + x^{\frac{1}{2}}, \quad du = \frac{1}{2} x^{-\frac{1}{2}} dx = \frac{1}{2} \frac{1}{\sqrt{x}} dx$$

$$2 du = \frac{1}{\sqrt{x}} dx \quad \checkmark \checkmark \checkmark \checkmark$$

$$\int \frac{2 du}{u^2} = 2 \frac{u^{-1}}{-1} = -2 \left[\frac{1}{1+\sqrt{x}} \right]_1^4 = -2 \left[\frac{1}{4} - \frac{1}{2} \right]$$

$$= -2 \left(-\frac{1}{4} \right) = \frac{1}{2} \quad \checkmark \checkmark \checkmark \checkmark \checkmark$$

[10]

UNIVERSITY OF BOTSWANA
DEPARTMENT OF MATHEMATICS
2019/2020 SUPPLEMENTARY EXAMINATIONS

FRONT PAGE

COURSE NUMBER MAT122 DURATION 2 hrs DATE JAN 2020
TITLE OF PAPER INTRODUCTORY MATHEMATICS II
SUBJECT MATHEMATICS TITLE OF EXAMINATION BSc I

MORNING/AFTERNOON

INSTRUCTIONS:

- ANSWER ALL QUESTIONS IN SECTION A AND ANY TWO QUESTIONS FROM SECTION B.
- ALL MARKS ARE INDICATED IN [].

NUMBER OF PAGES: 5

**DO NOT OPEN THIS PAGE UNTIL YOU HAVE BEEN TOLD TO DO
SO BY THE SUPERVISOR.**

SECTION A

Answer All Questions: Each Question carries 10 Marks

QUESTION 1

- (a) Find an equation of the conic section with vertices $(2, 1)$ and $(6, 1)$ and foci $(0, 1)$ and $(8, 1)$. [5]
- (b) Find the slope of the line tangent to $y = x \ln x$ at $x = 1$. [5]

QUESTION 2

Find the limits if they exist:

- (a) $\lim_{h \rightarrow 0} \frac{\sqrt{h^2 + 4h + 5} - \sqrt{5}}{h}$, [5]
- (b) $\lim_{x \rightarrow 0} \frac{3x}{\sin 2x}$, [5]

QUESTION 3

Find derivatives of the following functions.

- (a) $f(x) = 2x \sin x + x^2 \cos x$ [4]
- (b) $g(x) = \sqrt[3]{6x^2 + 1}$ [3]
- (c) $h(x) = e^{x^3} + \ln x - \frac{1}{x^3}$ [3]

QUESTION 4

- (a) Using the limit process, find the derivative of the function

$$f(x) = \frac{1}{\sqrt{x}}. \quad [5]$$

- (b) Evaluate $\frac{dy}{dx}$ at the point $(1, -4)$ where $y = (x - 2)(x^2 + 2x)$. [5]

QUESTION 5

- (a) Find the area of the region bounded by the graphs of the equations $y = 5x^2 + 2$, $x = 0$, $x = 2$ and $y = 0$. [5]

- (b) Show that $f(x) = x^3 + x^4 \sin x$ is an odd function.

$$\text{Hence evaluate } \int_{-1}^1 (x^3 + x^4 \sin x) dx \quad [5]$$

QUESTION 6

- (a) Solve for x leaving your answer in logarithmic form:

$$8e^x - 12 = 9. \quad [5]$$

- (b) Evaluate $\int_0^1 e^x (e^x + 1)^2 dx$. [5]

SECTION B

Answer TWO Questions OUT of FOUR:

Each Question carries 20 Marks

QUESTION 7

- (a) Write the definition of a continuous function $f(x)$ at a point $x = a$. [2]
- (b) Sketch The graph of $5x^2 + y^2 + 4y = 1$ [12]
- (c) Find a such that $f(x) = \begin{cases} 3ax^2, & x \geq 1 \\ 5 + x & x < 1 \end{cases}$ is continuous on the real line. [6]

QUESTION 8

Find the following integrals:

- (a) $\int (x^4 - 4^x) dx$. [3]
- (b) $\int \frac{\cos x}{1 - \sin x} dx$. [4]
- (c) $\int x^3 \ln x dx$. [5]
- (d) $\int \frac{dx}{x^2 - x - 6}$. [8]

QUESTION 9

(a) Differentiate

(i) $x^3y^3 - y = x$ [5]

(ii) $g(x) = \tan x^2 + \frac{e^x}{x}$ [4]

(b) Find an equation of the tangent line to the graph $f(x) = \frac{x+4}{2x-5}$ at the point $(9, 1)$. [6]

(c) Use logarithmic differentiation to find $\frac{dy}{dx}$ if $y = \frac{x^2\sqrt{3x-2}}{(x-1)^2}$, $x > \frac{3}{2}$. [7]

QUESTION 10

Consider the function $f(x) = \frac{x^2+1}{x-1}$ on $[0, 2]$.

- (a) Find the critical numbers of f . [3]
- (b) Find open intervals on which f is increasing or decreasing. [4]
- (c) Find the horizontal asymptotes and vertical asymptotes. [3]
- (d) Identify where the extrema of f occur. [4]
- (e) Find open intervals on which f is concave upward or concave downward. [6]

SOLNS

QUESTION 1

- (a) Find an equation of the conic section with vertices (2, 1) and (6, 1) and foci (0, 1) and (8, 1). [5]
(b) Find the slope of the line tangent to $y = x \ln x$ at $x = 1$. [5]

QUESTION 2 Find the limits if they exist:

- (a) $\lim_{h \rightarrow 0} \frac{\sqrt{h^2 + 4h + 5} - \sqrt{5}}{h}$. [5]
(b) $\lim_{x \rightarrow 0} \frac{3x}{\sin 2x}$. [5]

QUESTION 8

- (1) Write the definition of a continuous function $f(x)$ at a point $x = a$. [2]
(2) Sketch The graph of $5x^2 + y^2 + 4y = 1$ [12]
(3) Find a such that $f(x) = \begin{cases} 3ax^2, & x \geq 1 \\ 5 + x & x < 1 \end{cases}$ is continuous on the real line. [6]

SOLUTIONS

QUESTION 1

- (a) The conic section with vertices at (2, 1) and (6, 1) and foci (0, 1) and (8, 1) has a center at (4, 1) (and axis $y = 1$). [1]
The center to vertex distance is $a = 2$ and the center to focus distance is $c = 3$. [2]
The standard form of its equation is $\frac{(x-4)^2}{4} - \frac{(y-1)^2}{5} = 1$. [2]
(b) We have $y' = \ln x + 1$. [3]
The slope of the line tangent to $y = x \ln x$ at $x = 1$ is given by the value of its derivative at 1, i.e. $y'(1) = \ln 1 + 1 = 1$. [2]

QUESTION 2 Find the limits if they exist:

(a)

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{\sqrt{h^2 + 4h + 5} - \sqrt{5}}{h} &= \lim_{h \rightarrow 0} \frac{\sqrt{h^2 + 4h + 5} - \sqrt{5}}{h} \times \frac{\sqrt{h^2 + 4h + 5} + \sqrt{5}}{\sqrt{h^2 + 4h + 5} + \sqrt{5}} \\ &= \lim_{h \rightarrow 0} \frac{h^2 + 4h}{h(\sqrt{h^2 + 4h + 5} + \sqrt{5})} = \lim_{h \rightarrow 0} \frac{h + 4}{\sqrt{h^2 + 4h + 5} + \sqrt{5}} = \frac{2}{\sqrt{5}} \end{aligned}$$

$$(b) \lim_{x \rightarrow 0} \frac{3x}{\sin 2x} = \lim_{x \rightarrow 0} \frac{3x}{\sin 2x} = \lim_{x \rightarrow 0} \frac{3x}{2x} \times \lim_{x \rightarrow 0} \frac{1}{(\sin 2x)/2x} = \frac{3}{2}$$

QUESTION 7

- (a) A function $f(x)$ is continuous at a point $x = a$ if
- $f(a)$ is defined,

- $\lim_{x \rightarrow a} f(x)$ exists,
- $\lim_{x \rightarrow a} f(x) = f(a)$.

[1]
[1]

- (b) Completing the square in $5x^2 + y^2 + 4y = 1$ is obtain by :

$$5x^2 + y^2 + 4y + 4 = 1 + 4$$

$$5x^2 + (y + 2)^2 = 5$$

[2]

The standard form of the equation of the conic $5x^2 + y^2 + 4y = 1$ is $x^2 + \frac{(y+2)^2}{5} = 1$.

[1]

The conic is an ellipse with center at $(0, -2)$.

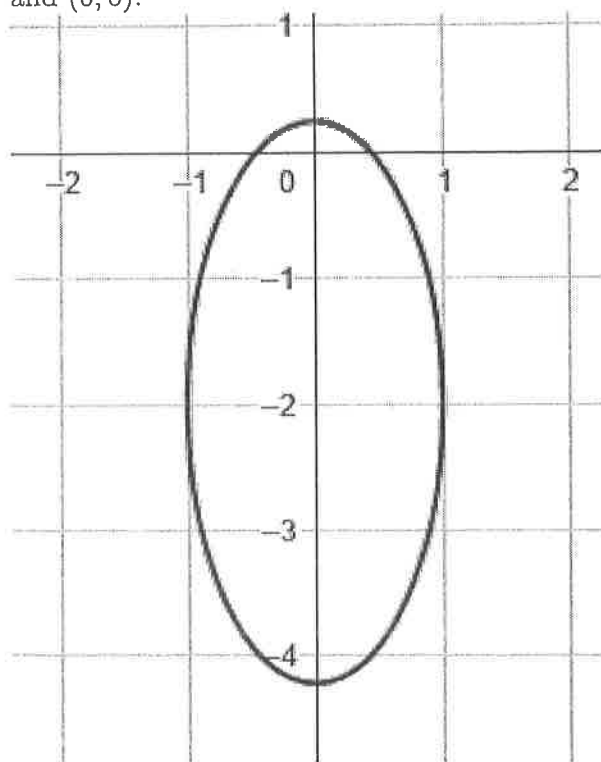
[1]

The half-length of the major axis is $a = \sqrt{5}$; it has vertices at $(0, -2 - \sqrt{5})$ and $(0, -2 + \sqrt{5})$

[2]

The half-length of the minor axis is $b = 1$. Hence $c = \sqrt{5 - 1} = 2$. Thus the foci are at $(0, -4)$ and $(0, 0)$.

[2]



[3]

- (c) The function $f(x) = \begin{cases} 3ax^2, & x \geq 1 \\ 5 + x & x < 1 \end{cases}$ is continuous at $x < 1$ and $x > 1$.

[2]

For f to be continuous on the real line it must also be continuous at $x = 1$.

We have $f(1) = 3a$, $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (5 + x) = 6$ and $\lim_{x \rightarrow 1^+} (3ax^2) = 3a[1] + [1] + [1]$

[1]

f is continuous at $x = 1$ if $6 = 3a$, i.e. $a = 2$.

Question 3

$$a) f'(x) = 2 \sin x + 2x \cos x + 2x \cos x - x^2 \sin x$$

$$b) g(x) = \sqrt[3]{6x^2+1} \\ = (6x^2+1)^{1/3}$$

$$\Rightarrow g'(x) = \frac{1}{3} (6x^2+1)^{-2/3} (12x)$$

$$= 2x(6x^2+1)^{-2/3}$$

$$c) h(x) = e^{x^3} + \ln x - \frac{1}{x^3}$$

$$\Rightarrow h'(x) = 2x^2 e^{x^3} + \frac{1}{x} + \frac{3}{x^2}$$

Question 4

$$a) f(x) = \frac{1}{\sqrt{x}}$$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{\frac{1}{\sqrt{x+\Delta x}} - \frac{1}{\sqrt{x}}}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\left(\frac{1}{\sqrt{x+\Delta x}} - \frac{1}{\sqrt{x}} \right) \left(\frac{1}{\sqrt{x+\Delta x}} + \frac{1}{\sqrt{x}} \right)}{\Delta x \left(\frac{1}{\sqrt{x+\Delta x}} + \frac{1}{\sqrt{x}} \right)}$$

$$= \lim_{\Delta x \rightarrow 0} \left(\frac{1}{x+\Delta x} - \frac{1}{x} \right) \checkmark$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\Delta x \left(\frac{1}{\sqrt{x+\Delta x}} + \frac{1}{\sqrt{x}} \right)}{x - (x+\Delta x)} \checkmark$$

$$= \lim_{\Delta x \rightarrow 0} \frac{-\Delta x}{\Delta x \left(\frac{1}{\sqrt{x+\Delta x}} + \frac{1}{\sqrt{x}} \right)} \checkmark$$

$$= \lim_{\Delta x \rightarrow 0} \frac{-1}{(x+\Delta x) \left(\frac{1}{\sqrt{x+\Delta x}} + \frac{1}{\sqrt{x}} \right)} \checkmark$$

$$= \frac{-1}{x \left(\frac{1}{\sqrt{x}} + \frac{1}{\sqrt{x}} \right)} = \frac{-1}{2\sqrt{x}}$$

$$b) \quad y = (x-2)(x^2+2x) \checkmark$$

$$\Rightarrow y' = x^2+2x + (x-2)(2x+2) \checkmark \checkmark$$

$$\Rightarrow \frac{dy}{dx} \Big|_{(1,-4)} = 1^2+2(1) + (1-2)(2+2) \checkmark$$

$$= 3 + (-1)(4) = -1 \checkmark$$

Q5 (a)

$$A = \int_0^2 (5x^2 + 2) dx = \left(\frac{5x^3}{3} + 2x \right) \Big|_0^2 \\ = \left(\frac{5(8)}{3} + 2(2) \right) = \frac{40+12}{3} = \frac{52}{3}$$

$$(b) \quad f(-x) = +(-x)^3 + (-x)^4 \sin(-x) \\ = -x^3 - x^4 \sin x = -f(x)$$

$$\therefore \int_{-1}^1 (x^3 + x^4 \sin x) dx = 0$$

$$6 (a) \quad 8e^x - 12 = 9 \Rightarrow 8e^x = 21 \\ \Rightarrow \ln(8e^x) = \ln 21 \Rightarrow \ln 8 + \ln e^x = \ln 21 \\ \Rightarrow x = \ln 21 - \ln 8 \Rightarrow x = \ln\left(\frac{21}{8}\right)$$

$$(b) \quad \int_0^1 e^x (e^x + 1)^2 dx$$

$$\text{let } u = e^x + 1, \quad du = e^x dx, \quad u(1) = e+1, \quad u(0) = 2$$

$$\therefore \int_2^{e+1} u^2 du \Rightarrow \left[\frac{u^3}{3} \right]_2^{e+1}$$

$$= \frac{(e+1)^3}{3} - \frac{8}{3} = \frac{(e+1)^3 - 8}{3}$$

$$8(a) \int (x^4 - 4^x) dx$$

$$= \frac{x^5}{5} - \frac{4^x}{\ln 4} + C \quad \checkmark \checkmark$$

$$(b) \int \frac{\cos x}{1 - \sin x} dx \quad \text{let } u = 1 - \sin x, \quad du = -\cos x dx \quad \checkmark$$

$$\therefore - \int \frac{du}{u} = -\ln|u| + C = -\ln|1 - \sin x| + C \quad \checkmark$$

$$(c) \int x^3 \ln x dx, \quad \text{let } u = \ln x, \quad dv = x^3 dx$$

$$du = \frac{dx}{x} \quad v = \frac{x^4}{4}$$

$$I = \frac{x^4 \ln x}{4} - \int \frac{x^4}{4} \cdot \frac{dx}{x} = \frac{x^4 \ln x}{4} - \int \frac{x^3}{4} dx$$

$$= \frac{x^4 \ln x}{4} - \frac{x^4}{12} + C.$$

$$(d) \quad x^2 - x - 6 = (x-3)(x+2) \quad \checkmark, \quad \frac{1}{x^2 - x - 6} = \frac{A}{x-3} + \frac{B}{x+2} \quad \checkmark$$

$$\Rightarrow 1 = A(x+2) + B(x-3) \Rightarrow 1 = -5B \Rightarrow B = -\frac{1}{5} \quad \checkmark$$

$$\Rightarrow 1 = 5A \Rightarrow A = \frac{1}{5} \quad \checkmark$$

$$\therefore \int \frac{dx}{x^2 - x - 6} = \frac{1}{5} \int \frac{dx}{x-3} - \frac{1}{5} \int \frac{dx}{x+2} \quad \checkmark \checkmark$$

$$= \frac{1}{5} \ln|x-3| - \frac{1}{5} \ln|x+2| + C. \quad \checkmark \checkmark$$

Question 9

a)

(i) $x^3 y^3 - y = x$

$$3x^2 y^3 + 3x^3 y^2 \frac{dy}{dx} - \frac{dy}{dx} = 1 \quad \checkmark \checkmark$$

$$\Rightarrow (3x^3 y^2 - 1) \frac{dy}{dx} = 1 - 3x^2 y^3 \quad \checkmark$$

$$\Rightarrow \frac{dy}{dx} = \frac{1 - 3x^2 y^3}{3x^3 y^2 - 1} \quad \checkmark$$

(ii) $g(x) = \tan(x^2) + \frac{e^x}{x}$

$$\begin{aligned} \Rightarrow g'(x) &= 2x \sec^2(x^2) + \frac{x e^x - e^x}{x^2} \\ &= 2x \sec^2(x^2) + \frac{(x-1)e^x}{x^2} \quad \checkmark \end{aligned}$$

b) $f(x) = \frac{2x+4}{2x-5}$

$$\Rightarrow f'(x) = \frac{2x-5 - 2(x+4)}{(2x-5)^2} \quad \checkmark$$

$$= \frac{2x-5-2x-8}{(2x-5)^2} \quad \checkmark$$

$$= -\frac{13}{(2x-5)^2} \quad \checkmark$$

∴ The slope is $m = -\frac{13}{(2-5)^2} = -\frac{13}{9} \quad \checkmark$

and so the eqn of the tgl line is
 $y-9 = -\frac{13}{9}(x-1) \quad \checkmark$

$$c) \quad y = \frac{x^2 \sqrt{3x-2}}{(x-1)^2}$$

$$\Rightarrow \ln y = \ln \frac{x^2 \sqrt{3x-2}}{(x-1)^2} \quad \checkmark$$

$$= \ln x^2 + \frac{1}{2} \ln(3x-2) - 2 \ln(x-1) \quad \checkmark \checkmark$$

$$\Rightarrow \frac{1}{y} y' = \frac{2x}{x^2} + \frac{1}{2} \frac{3}{3x-2} - \frac{2}{x-1}$$

$$\Rightarrow y' = \left[\frac{2x}{x^2} + \frac{1}{2} \frac{3}{3x-2} - \frac{2}{x-1} \right] \frac{x^2 \sqrt{3x-2}}{(x-1)^2} \quad \checkmark \checkmark \checkmark$$

Question 10

$$f(x) = \frac{x^2 + 1}{x - 1}$$

$$a) f'(x) = \frac{2x(x-1) - (x^2+1)}{(x-1)^2} = \frac{2x^2 - 2x - x^2 - 1}{(x-1)^2} = \frac{x^2 - 2x - 1}{x-1}$$

$$f'(x) = 0 \Rightarrow \text{none}$$

$f'(x)$ is undefined when $x-1=0 \Rightarrow x=1$

$\therefore$ crit no : $x=1$ ✓

b) Horiz asy : $x-1=0 \Rightarrow x=1$ ✓

Vertical asy :

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{x^2 + 1}{x-1} = \frac{\infty}{\infty} = \infty$$

$$\lim_{x \rightarrow -\infty} f(x) = \lim_{x \rightarrow -\infty} \frac{-1 - 1/x^2}{-1/x + 1/x^2} = \frac{-1}{0} = -\infty$$

Non

c) ~~test~~ $[0, 1)$ $(1, 2]$

Test value

$f'(x)$

$$f'(0) = \frac{-1}{1} = -1 > 0$$

$$f'(2) = \frac{4 - 4 - 1}{1} = -1 < 0$$

Conclusion

incr

decr

d)

x
 $f(x)$

0

-1

min

undef

2

5

max

$$e) \quad f''(x) = \frac{(2x-2)(x-1) - (x^2-2x+1)}{(x-1)^2}$$

$$= \frac{2x^2 - 4x + 2 - x^2 + 2x + 1}{(x-1)^2}$$

$$= \frac{x^2 - 2x + 3}{(x-1)^2}$$

$$x^2 - 2x + 3 = 0$$

$$\Rightarrow x = \frac{2 \pm \sqrt{4-12}}{2}$$

$$f''(x) = 0 \Rightarrow \text{none}$$

$f''(x)$ is undefined when $x-1=0 \Rightarrow x=1$.

$[0, 1)$

$(1, 2]$

Test
value

0

2

$f''(x)$

$$f''(0) = \frac{3}{1} = 3 > 0$$

$$f''(2) = \frac{4-4+3}{1} = 3 > 0$$

Conclusion

CU

CU



UNIVERSITY OF BOTSWANA
DEPARTMENT OF MATHEMATICS
2019/2020 –FINAL EXAMINATIONS

FRONT PAGE

COURSE NUMBER MAT122 DURATION 2 hrs DATE NOV 2019
TITLE OF PAPER INTRODUCTORY MATHEMATICS II
SUBJECT MATHEMATICS TITLE OF EXAMINATION BSc I

MORNING/AFTERNOON

INSTRUCTIONS:

- ANSWER ALL QUESTIONS IN SECTION A AND ANY TWO QUESTIONS FROM SECTION B.
- ALL MARKS ARE INDICATED IN [].

NUMBER OF PAGES: 5

**DO NOT OPEN THIS PAGE UNTIL YOU HAVE BEEN TOLD TO DO
SO BY THE SUPERVISOR.**

SECTION A

Answer All Questions: Each Question carries 10 Marks

Question 1

- (a) Find an equation of the conic section with vertices $(1, 2)$ and $(1, 6)$ and foci $(1, 0)$ and $(1, 8)$. [5]
- (b) Find the slope of the line tangent to $y = x \sin x$ at $x = \frac{\pi}{2}$. [5]

Question 2

Find the limits if they exist:

- (a) $\lim_{h \rightarrow 0} \frac{\sqrt{h^2 + 4h + 3} - \sqrt{3}}{h}$. [5]
- (b) $\lim_{x \rightarrow 0} \frac{\tan 3x}{2x}$. [5]

Question 3

Find derivatives of the following functions.

- (a) $f(x) = \frac{2}{\sqrt[3]{x}} + 3 \tan x$ [4]
- (b) $g(x) = (9x + 2)^{\frac{2}{3}}$ [3]
- (c) $h(x) = e^{x^2} - \frac{1}{x}$ [3]

Question 4

- (a) Using the limit process, find the derivative of the function $f(x) = 2x^2$. [5]
- (b) Evaluate $\frac{dy}{dx}$ at the point $(0, \frac{\pi}{2})$ where $\cos(x + y) = x$. [5]

Question 5

- (a) Find $\int \left(x^5 - \sin x + \frac{1}{x} \right) dx$. [4]
- (b) Show that the area bounded by the curves $y = \frac{x}{x^2 + 1}$, $x = 0$ and $x = 3$ is $\frac{1}{2} \ln 10$. [6]

Question 6

- (a) Evaluate $\int_0^1 x(x + 1)^3 dx$. [6]
- (b) Given that $\int_0^5 f(x) dx = 10$ and $\int_5^7 f(x) dx = -1$, evaluate $\int_0^7 f(x) dx$. [4]

SECTION B

**Answer TWO Questions OUT of FOUR:
Each Question carries 20 Marks**

Question 7

- (a) Define what is meant by a function $f(x)$ being continuous at a point $x = a$. [3]
- (b) Find a such that $f(x) = \begin{cases} 1 - 3ax^2, & x \geq 1 \\ 5 + x & x < 1 \end{cases}$ is continuous on the real line. [6]
- (c) Find the center, vertices, foci of the conic section $x^2 + 5y^2 + 4x = 1$ and sketch its graph. [11]

Question 8

- (a) Find $\int \tan x dx$. [5]
- (b) Find $\int e^x \sin x dx$. [7]
- (c) Find the area bounded by the curves $y = 4 - x^2$ and $y = x^2 - 4$. [8]

Question 9

(a) Find derivatives of the following functions.

(i) $f(x) = (x^3 + 2x^2 - x)^3$ [4]

(ii) $g(x) = \sin 2x + xe^x$ [5]

(b) Find an equation of the tangent line to the graph $f(x) = (4x + 1)^2$ at the point $(0, 1)$. [6]

(c) Use logarithmic differentiation to find $\frac{dy}{dx}$ if $y = \sqrt{\frac{x^2 - 1}{x^2 + 1}}$, $x > 1$. [7]

Question 10

Consider the function $f(x) = x^4 - 4x^3 + 2$ on $[-1, 4]$.

(a) Find the critical numbers of f . [3]

(b) Find open intervals on which f is increasing or decreasing. [6]

(c) Identify where the extrema of f occur. [4]

(d) Find open intervals on which f is concave upward or concave downward. [4]

(e) Sketch the curve of f . [3]

QUESTION 1

(a) The conic section with vertices at (1, 2) and (1, 6) and foci at (1, 1) and (1, 7) has a center at (1, 4) (and axis $x = 1$). [1]

The center to vertex distance is $a = 2$ and the center to focus distance is $c = 3$. [2]

The standard form of its equation is $\frac{(y-4)^2}{4} - \frac{(x-1)^2}{5} = 1$. [2]

(b) We have $y' = \sin x + x \cos x$. [3]

The slope of the line tangent to $y = x \sin x$ at $x = \frac{\pi}{2}$ is given by the value of its derivative at $\frac{\pi}{2}$, [2]

i.e. $y' \left(\frac{\pi}{2} \right) = \sin \frac{\pi}{2} + \frac{\pi}{2} \cos \frac{\pi}{2} = 1$.

QUESTION 2 Find the limits if they exist:

(a)

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{\sqrt{h^2 + 4h + 3} - \sqrt{3}}{h} &= \lim_{h \rightarrow 0} \frac{\sqrt{h^2 + 4h + 3} - \sqrt{3}}{h} \times \frac{\sqrt{h^2 + 4h + 3} + \sqrt{3}}{\sqrt{h^2 + 4h + 3} + \sqrt{3}} \\ &= \lim_{h \rightarrow 0} \frac{h^2 + 4h}{h(\sqrt{h^2 + 4h + 3} + \sqrt{3})} = \lim_{h \rightarrow 0} \frac{h + 4}{\sqrt{h^2 + 4h + 3} + \sqrt{3}} = \frac{2}{\sqrt{3}} \end{aligned} \quad [5]$$

$$(b) \lim_{x \rightarrow 0} \frac{\tan 3x}{2x} = \lim_{x \rightarrow 0} \frac{\sin 3x}{2x \cos 3x} = \lim_{x \rightarrow 0} \frac{\sin 3x}{2x} \times \lim_{x \rightarrow 0} \frac{1}{\cos 3x} = \frac{3}{2} \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \times \lim_{x \rightarrow 0} \frac{1}{\cos 3x} = \frac{3}{2}. \quad [5]$$

QUESTION 7

- (a) A function $f(x)$ is continuous at a point $x = a$ if
 - $f(a)$ is defined,
 - $\lim_{x \rightarrow a} f(x)$ exists,
 - $\lim_{x \rightarrow a} f(x) = f(a)$.
- (b) Completing the square in $x^2 + 5y^2 + 4x = 1$ is obtain by :

$$x^2 + 4x + 4 + 5y^2 = 1 + 4$$

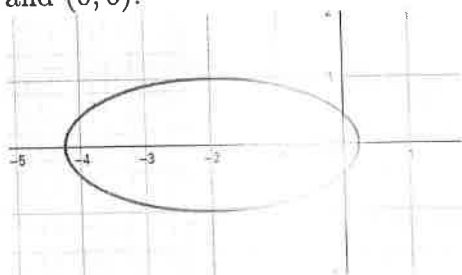
$$(x + 2)^2 + 5y^2 = 5$$

The standard form of the equation of the conic $x^2 + 5y^2 + 4x = 1$ is $\frac{(x+2)^2}{5} + y^2 = 1$. [2]

The conic is an ellipse with center at $(-2, 0)$. [1]

The half-length of the major axis is $a = \sqrt{5}$; it has vertices at $(-2 - \sqrt{5}, 0)$ and $(-2 + \sqrt{5}, 0)$ [1]

The half-length of the minor axis is $b = 1$. Hence $c = \sqrt{5 - 1} = 2$. Thus the foci are at $(-4, 0)$ and $(0, 0)$. [2]



- (c) The function $f(x) = \begin{cases} 1 - 3ax^2, & x \geq 1 \\ 5 + x & x < 1 \end{cases}$ is continuous at $x < 1$ and $x > 1$. [2]

For f to be continuous on the real line it must also be continuous at $x = 1$.

We have $f(1) = 1 - 3a$, $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (5 + x) = 6$ and $\lim_{x \rightarrow 1^+} (1 - 3ax^2) = 1 - 3a$
 [1]+[1]+[1]
 f is continuous at $x = 1$ if $6 = 1 - 3a$, i.e. $a = -5/3$. [1]

Question 3

$$a) f(x) = 2x^{-1/3} + 3 \sec^2 x \tan x$$
$$\Rightarrow f'(x) = -\frac{2}{3} x^{-4/3} - 3 \sec^2 x$$

$$b) g'(x) = \frac{2}{3} (9x+2)^{-1/3} (9)$$
$$= 6 (9x+2)^{-1/3}$$
$$= \frac{6}{(9x+2)^{1/3}}$$

$$c) h'(x) = 2xe^{x^2} + \frac{1}{x^2}$$

Question 4

$$a) f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$
$$= \lim_{\Delta x \rightarrow 0} \frac{\frac{2}{\sqrt{x+\Delta x}} - \frac{2}{\sqrt{x}}}{\Delta x}$$
$$= 2 \lim_{\Delta x \rightarrow 0} \frac{\frac{1}{\sqrt{x+\Delta x}} - \frac{1}{\sqrt{x}}}{\Delta x}$$
$$= 2 \lim_{\Delta x \rightarrow 0} \left(\frac{\frac{1}{\sqrt{x+\Delta x}} - \frac{1}{\sqrt{x}}}{\Delta x} \right) \left(\frac{1}{\sqrt{x+\Delta x}} + \frac{1}{\sqrt{x}} \right)$$
$$= 2 \lim_{\Delta x \rightarrow 0} \frac{\left(\frac{1}{\sqrt{x+\Delta x}} - \frac{1}{\sqrt{x}} \right)}{\Delta x \left(\frac{1}{\sqrt{x+\Delta x}} + \frac{1}{\sqrt{x}} \right)} = 2 \lim_{\Delta x \rightarrow 0} \frac{x - (x+\Delta x)}{\Delta x (x+\Delta x) x \left(\frac{1}{\sqrt{x+\Delta x}} + \frac{1}{\sqrt{x}} \right)}$$

$$= 2 \lim_{\Delta x \rightarrow 0} \frac{-1}{x(x+\Delta x) \left(\frac{1}{\sqrt{x+\Delta x}} + \frac{1}{\sqrt{x}} \right)}$$

$=$

Question 4

a) $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$

$$= \lim_{\Delta x \rightarrow 0} \frac{2(x+\Delta x)^2 - 2x^2}{\Delta x}$$

$$= 2 \lim_{\Delta x \rightarrow 0} \frac{(x^2 + 2x\Delta x + \Delta x^2 - x^2)}{\Delta x}$$

$$= 2 \lim_{\Delta x \rightarrow 0} \frac{(2x + \Delta x) \cancel{\Delta x}}{\cancel{\Delta x}}$$

$$= 2(2x) = 4x$$

b) $\frac{\cos}{\cancel{\tan}}(x+y) = x$

$$\Rightarrow \frac{-\sin(x+y)}{\sec^2(x+y)} \left(1 + \frac{dy}{dx} \right) = 1$$

$$\Rightarrow 1 + \frac{dy}{dx} = \frac{1}{\sec^2(x+y)} = \cos^2(x+y)$$

$$\Rightarrow \frac{dy}{dx} = \cos^2(x+y) - 1$$

$$\begin{aligned} \therefore \frac{dy}{dx} \Big|_{(0,0)} &= \cos^2(0+0) - 1 \\ &= \cos^2(0) - 1 = 1 - 1 = 0 \end{aligned}$$

$$b) \cos(x+y) = x$$

$$\Rightarrow -\sin(x+y) \left(1 + \frac{dy}{dx}\right) = 1$$

$$\Rightarrow 1 + \frac{dy}{dx} = \frac{-1}{\sin(x+y)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-1}{\sin(x+y)} - 1$$

$$\Rightarrow \frac{dy}{dx} \Big|_{(0, \frac{\pi}{2})} = \frac{-1}{\sin(\frac{\pi}{2})} - 1$$

$$= -1 - 1 = -2$$

Q5(a)

$$\int \left(x^5 - \sin x + \frac{1}{x} \right) dx$$

$$= \frac{x^6}{6} + \cos x + \ln |x| + C$$

6(a) $\int_0^1 x(x+1)^3 dx$

let $u = x+1$, $du = dx$ ✓

$$\int_0^1 x(x+1)^3 dx$$

$$= \int u^3 du = \frac{u^4}{4} + C$$

(b) $\int_0^3 \left(\frac{x}{x^2+1} \right) dx$

let $u = x^2+1$ ✓, $u = 1, 10$ ✓
 $du = 2x dx$
 $\frac{du}{2} = x dx$

$$\int \frac{x dx}{x^2+1} = \frac{1}{2} \int \frac{du}{u}$$

$$= \frac{1}{2} \ln |u| + C$$

$$\therefore \int_0^3 \frac{x}{x^2+1} dx =$$

$$\frac{1}{2} \ln |u| \Big|_1^{10}$$

$$= \frac{1}{2} [\ln 10 - \ln 1]$$

$$= \frac{1}{2} \ln 10$$

$$\Rightarrow \int_0^1 x(x+1)^3 dx$$

$$= \left. \frac{(x+1)^4}{4} \right|_0^1$$

$$= \frac{2^4}{4} - \frac{1}{4} = \frac{16}{4} - \frac{1}{4} = \frac{15}{4}$$

(b) $\int_0^7 f(x) dx = \int_5^7 f(x) dx + \int_0^5 f(x) dx$ ✓

$$= -1 - 10$$

$$= -11$$

Question 8

a) i) $f'(x) = 3(x^3 + 2x^2 - x)^2 (3x^2 + 4x - 1)$

ii) $g'(x) = 2\cos 2x + e^x + xe^x$

b) $f'(x) = 2(4x+1)(4) = 8(4x+1)$

The slope is
 $m = f'(0) = 8(4(0)+1) = 8$

∴ The eqⁿ of the tgt line is

$$y-1 = 8(x-0)$$

or $y = 8x+1$

c) $x^3 - xy + y^2 = 4$

$$\Rightarrow 3x^2 - y - x \frac{dy}{dx} + 2y \frac{dy}{dx} = 0$$

$$\Rightarrow (2y - x) \frac{dy}{dx} = y - 3x^2$$

$$\Rightarrow \frac{dy}{dx} = \frac{y - 3x^2}{2y - x}$$

$$c) \quad y = \sqrt{\frac{x^2-1}{x^2+1}}$$

$$\Rightarrow \ln y = \ln \left(\frac{x^2-1}{x^2+1} \right)^{\frac{1}{2}}$$

$$= \frac{1}{2} \ln \left(\frac{x^2-1}{x^2+1} \right)$$

$$= \frac{1}{2} \left[\ln(x^2-1) - \ln(x^2+1) \right]$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \frac{1}{2} \left[\frac{2x}{x^2-1} - \frac{2x}{x^2+1} \right]$$

$$\Rightarrow \frac{dy}{dx} = \cancel{x} \left[\frac{1}{x^2-1} - \frac{1}{x^2+1} \right] \sqrt{\frac{x^2-1}{x^2+1}}$$

Q(a)

$$\int \tan x \, dx$$

$$= \int \frac{\sin x}{\cos x} \, dx$$

let $u = \cos x$, $du = -\sin x \, dx$ ✓

$$\therefore \int \tan x \, dx = - \int \frac{-\sin x \, dx}{\cos x} \quad \checkmark$$

$$= - \int \frac{du}{u} = -\ln|u| + C \quad \checkmark$$

$$= -\ln|\cos x| + C \quad \checkmark$$

$$= \ln\left|\frac{1}{\cos x}\right| + C$$

$$= \ln|\sec x| + C \quad \checkmark$$

(b) $\int e^x \sin x \, dx$

let $e^x = u$, $du = e^x \, dx$ ✓
 $dv = \sin x \, dx$, $v = -\cos x$ ✓

$$I = -e^x \cos x - \int -\cos x \cdot e^x \, dx \quad \checkmark$$

$$= -e^x \cos x + \int e^x \cos x \, dx$$

let $u = e^x$, $du = e^x \, dx$ ✓
 $dv = \cos x \, dx$, $v = \sin x$ ✓

$$I = -e^x \cos x + e^x \sin x - \int e^x \sin x \, dx$$

$$I = -e^x \cos x + e^x \sin x - I \quad \checkmark$$

$$\Rightarrow 2I = +e^x \sin x - e^x \cos x \quad \checkmark$$

$$\Rightarrow I = \frac{e^x (\sin x - \cos x)}{2} + C \quad \checkmark$$

(c) $4 - x^2 = x^2 - 4$

$$\Rightarrow 2x^2 - 8 = 0$$

$$\Rightarrow x^2 - 4 = 0 \quad \checkmark$$

$$\Rightarrow x = \pm 2 \quad \checkmark$$

$$A = \int_{-2}^2 (4 - x^2 - x^2 + 4) \, dx \quad \checkmark$$

$$= \int_{-2}^2 (8 - 2x^2) \, dx \quad \checkmark$$

$$= 8x - \frac{2x^3}{3} \bigg|_{-2}^2 \quad \checkmark$$

$$= 8(2) - \frac{2(2)^3}{3} - 8(-2) + \frac{2(-2)^3}{3} \quad \checkmark$$

$$= 16 - \frac{16}{3} + 16 - \frac{16}{3} \quad \checkmark$$

$$= 32 - \frac{32}{3} = \frac{96 - 32}{3} = \frac{64}{3} \quad \checkmark$$

Question 10

$$f(x) = x^4 - 4x^3 + 2$$

a) crit no:

$$f'(x) = 4x^3 - 12x^2 = 0$$

$$\Rightarrow 4x^2(x-3) = 0$$

$$\Rightarrow x=0 \text{ or } x=3$$

$$27(3-4) + 2$$

$$= -27 + 2$$

$$64(4-4) + 2 = 2$$

b)

Test ints	Test values	$f'(x)$	Concl ⁿ
$(-1, 0)$	-1	$f'(-1) = -4 - 12 = -16 < 0$	decr
$(0, 3)$	1	$f'(1) = 4 - 12 = -8 < 0$	decr
$(3, 4)$	4	$f'(4) = 256 - 192 = 64 > 0$	incr

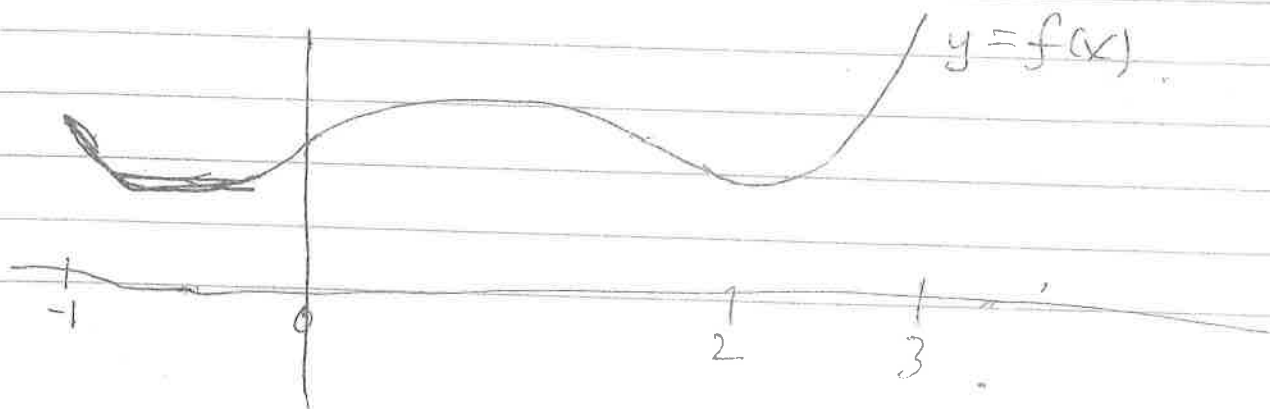
c)	x	-1	0	3	4
	f(x)	7	2	-25	2
		/		/	
		max		min	

d) inflexion pts:

$$f''(x) = 12x^2 - 24x = 12x(x-2) = 0$$

$$\Rightarrow x=0 \text{ or } x=2$$

Test ints	Test value	$f''(x)$	Conclusion
$(-1, 0)$	-1	$f''(-1) = -12(-3) = 36 > 0$	CU
$(0, 2)$	1	$f''(1) = 12(-1) = -12 < 0$	CD
$(2, 4)$	3	$f''(3) = 36(1) = 36 > 0$	CU



UNIVERSITY OF BOTSWANA

DEPARTMENT OF MATHEMATICS

Examination: Semester II

Year: 2018/19

COURSE CODE: **MAT 122** DURATION: **2HRS** DATE: **MAY 2019**

TITLE OF PAPER: **INTRODUCTORY MATHEMATICS II**

SUBJECT: **MATHEMATICS** TITLE OF EXAMINATION: **Bsc I**

INSTRUCTIONS:

- Answer all questions in SECTION A.
- In SECTION B answer any two (2) out of four (4) questions.

No. OF PAGES: 5

DO NOT OPEN THIS PAPER UNTIL YOU HAVE BEEN TOLD TO
DO SO BY THE ENVIGILATOR.

Section A

Answer All Questions: Each Question carries 10 Marks.

Question 1

- (a) Find an equation of the circle centered at $(-1, 3)$ and passing through the point $(1, 1)$. [5]
- (b) Find the vertex, focus and directrix of the parabola $(x + \frac{1}{2})^2 = 4(y - 1)$. [5]

Question 2

Evaluate the following limits.

- (a) $\lim_{x \rightarrow 0} \left(3x + \frac{\sin 4x}{x} \right)$ [3]
- (b) $\lim_{x \rightarrow -\infty} \frac{2x^2 + 3x - 1}{5x^2 + 1}$ [4]
- (c) $\lim_{x \rightarrow 2} \frac{x^2 + x - 6}{3x^2 - 12}$ [3]

Question 3

- (a) Find $\frac{dy}{dx}$ for the function $y = \frac{e^{2x}}{x^2 + 1}$. [5]
- (b) Determine $\frac{dy}{dx}$ given that $x + xy - y^3 = 7$. [5]

Question 4

- (a) Evaluate $\frac{dy}{dx}$ for the curve given by

$$x = 2 \cos \theta, \quad y = 2 \sin \theta$$

at the point for which $\theta = \pi/4$. [4]

- (b) Find the global extrema of the function

$$f(x) = x^2 - 3x^{2/3}$$

on the interval $[-1, 8]$. [6]

Question 5

Determine the following integrals;

(a) $\int (\sqrt{x} + \frac{1}{2x}) dx.$ [3]

(b) $\int_{-\pi/4}^{\pi/4} (\sec^2 t + \sin t) dt.$ [4]

(c) $\int \frac{2x}{(x^2 + 1)^{1/3}} dx.$ [3]

Question 6

(a) Find the area of the region bounded by $y = 1 + x^2$, the x -axis and the lines $x = -1$ and $x = 1$. [5]

(b) Evaluate;

$$\int_0^1 xe^x dx,$$

[5]

Section B

Answer TWO Questions out of FOUR:
Each Question carries 20 Marks.

Question 7

- (a) Consider the equation of an ellipse given by

$$9x^2 + 4y^2 - 54x + 40y + 37 = 0.$$

- (i) Show that the standard equation of this ellipse is

$$\frac{(x-3)^2}{16} + \frac{(y+5)^2}{36} = 1.$$

[3]

- (ii) Find the centre, vertices and the foci of the given ellipse, and hence sketch the curve. [9]

- (b) Describe the conic section

$$\frac{(x-1)^2}{9} - \frac{y^2}{4} = 1$$

stating all its main features.

[8]

Question 8

- (a) Write down the definition of a continuous function $f(x)$ at a point $x = c$. [3]

- (b) Find the value of a , for $f(x)$ to be continuous on the real line [7]

$$f(x) = \begin{cases} -3ax^2, & x \geq 1 \\ 5 + x, & x < 1. \end{cases}$$

- (c) Evaluate the limit: $\lim_{x \rightarrow 0} \frac{\sin 3x}{6 \sin 4x}$. [4]

- (d) Use the limit definition of the differentiation to find the derivative of $f(x) = \frac{1}{\sqrt{x}}$. [6]

Question 9

Consider the curve given by $f(x) = x^4 - 2x^2$.

- (a) Find the critical points of the curve. [4]
- (b) Determine the interval on which the graph of the curve is increasing or decreasing. [5]
- (c) Classify the critical points. [2]
- (d) Determine the interval for which the graph of the curve is concave upwards or downwards and the point of inflection. [5]
- (e) Hence sketch the graph of the function f . [4]

Question 10

- (a) Find $\int \frac{x}{x^2 - x - 12} dx$. [5]
- (b) Evaluate $\int_0^{\pi/2} x \sin x dx$. [5]
- (c) Use trigonometric substitution to find $\int \frac{dx}{\sqrt{9 - x^2}}$. [5]
- (d) Use logarithm to find $\frac{dy}{dx}$ given that $y = \sqrt[4]{(x^3 + 1)/x^{7/9}}$. [5]

DEPARTMENT OF MATHEMATICS

Course Moderation Form

ACADEMIC YEAR: 2018-19 **SEMESTER:** 1

COURSE CODE & TITLE: MAT122 - Introductory Mathematics II

LECTURER: Prof. Robdera

MODERATORS: Mr Ramatebele, Mr Moetele

Lecture Notes (reflect standards/course outline) _____

Assignments (general comments) _____

Assignment/Test (general comments): _____

Test I (seen/signed/dated): _____

Test II (seen/signed/dated): _____

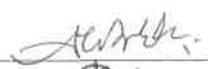
C.A. based on: 3 tests

Final Exam: All tests seen ~~now~~

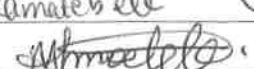
Overall agreement for the course between Lecturer and Moderator:

Accepted.

SIGNATURES

Lecturer: ROBDERA 

Moderator I: Ramatebele 

Moderator II: Moetele 

Date: _____

Date: 25/10/18

Date: 30/10/18

UNIVERSITY OF BOTSWANA
DEPARTMENT OF MATHEMATICS
2018/2019 – EXAMINATIONS

FRONT PAGE

COURSE NUMBER MAT122 **DURATION** 2 hrs **DATE** DEC 2018
TITLE OF PAPER INTRODUCTORY MATHEMATICS II
SUBJECT MATHEMATICS **TITLE OF EXAMINATION** BSc I

MORNING/AFTERNOON

INSTRUCTIONS:

- ANSWER ALL QUESTIONS IN SECTION A AND ANY TWO QUESTIONS FROM SECTION B.
- ALL MARKS ARE INDICATED IN [].

NUMBER OF PAGES: 5

**DO NOT OPEN THIS PAGE UNTIL YOU HAVE BEEN TOLD TO DO
SO BY THE SUPERVISOR.**

SECTION A

Answer All Questions: Each Question carries 10 Marks.

Question 1

- (a) Find the standard form of the equation of the circle with center at $(3, 7)$ and passes through $(1, 0)$. [5]

- (b) Find the center of the ellipse whose equation is

$$x^2 - 6x + 2y^2 + 8y = -13.$$

[5]

Question 2

- (a) Given that $\sin \theta = \frac{1}{2}$ find the value of θ in radians for
 $0 < \theta < \frac{\pi}{2}$. [3]

- (b) Verify the identity $\frac{\tan x \cot x}{\cos x} = \sec x$. [3]

- (c) Find the exact value of $\arctan \left(\frac{\sqrt{3}}{3} \right)$. [4]

Question 3

(a) Find the limits if they exist:

(i) $\lim_{x \rightarrow 0} (2x + \frac{\sin 2x}{x})$. [3]

(ii) $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$. [3]

(b) Find a such that $f(x) = \begin{cases} -3ax^2, & x \geq 2 \\ 5 + x & x < 2 \end{cases}$ is continuous on the real line. [4]

Question 4

(a) Find the integral $\int \frac{1+x}{\sqrt{x}} dx$. [5]

(b) Find the integral $\int \sin x \cos x dx$. [5]

Question 5

(a) Find the derivative of $y = \frac{2x+1}{x+3}$. [3]

(b) Find the derivative of $y = \sin x - x \cos x$. [3]

(c) Find the derivative of $y = 5 \ln x^4$. [2]

(d) Find the derivative of $y = xe^x$. [2]

Question 6

Find the extrema of $f(x) = 2x^3 - 6x$ on the closed interval $[0, 3]$. [10]

SECTION B

**Answer TWO Questions OUT of FOUR:
Each Question carries 20 Marks.**

Question 7

Consider the curve $x^2y^2 = 9$.

- (a) Show that the point $(3, -1)$ lies on the curve. [2]**
- (b) Find $\frac{dy}{dx}$. [6]**
- (c) Find the slope of the curve at the point $(1, 3)$. [2]**
- (d) Find the tangent line and the normal line to the curve at $(-1, 3)$. [10]**

Question 8

Consider

$$f(x) = 3x^4 - 4x^3 + 2.$$

- (a) Find the critical numbers of f . [4]**
- (a) Find the open intervals on which f is increasing or decreasing. [6]**
- (b) Identify where the extrema of f occur. [1]**
- (c) Find the intervals on which f is concave up and where it is concave down. [7]**
- (d) Find all points of inflection of f . [2]**

Question 9

(a) Find the following integrals

(i) $\int \frac{1}{\sqrt{x}(1+\sqrt{x})^2} dx.$ [7]

(ii) $\int e^{\sec 2x} \sec 2x \tan 2x dx.$ [7]

(b) Show that $\int_0^{\frac{\pi}{3}} \frac{\sin x}{(1+\cos x)^2} dx = \frac{1}{6}.$ [6]

Question 10

(a) Find the vertex, focus and directrix of the conic section

$$y^2 = -6x.$$

[4]

(b) Solve $\sin 2x \sin x = 2 \cos x$ in the interval $[0, 2\pi)$. [9]

(c) Find the standard form of the equation, center and vertices of the hyperbola given by

$$x^2 - 9y^2 + 36y - 72 = 0.$$

[7]

$$1 \quad (a) \quad (x-3)^2 + (y-7)^2 = (1-3)^2 + 7^2$$

$$\Rightarrow (x-3)^2 + (y-7)^2 = 4 + 49 \quad \checkmark$$

$$\Rightarrow (x-3)^2 + (y-7)^2 = 53 \quad \checkmark$$

$$b) \quad x^2 - 6x + 2y^2 + 8y = -13$$

$$\Rightarrow (x-3)^2 - 9 + 2(y^2 + 4y) = -13 \quad \checkmark$$

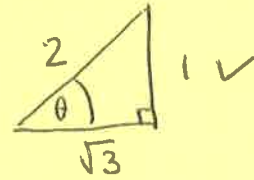
$$\Rightarrow (x-3)^2 - 9 + 2(y+2)^2 - 8 = -13 \quad \checkmark$$

$$\Rightarrow (x-3)^2 + 2(y+2)^2 = 4 \quad \checkmark$$

$$\Rightarrow \frac{(x-3)^2}{4} + \frac{(y+2)^2}{2} = 1 \quad \checkmark$$

$$c(3, -2) \quad \checkmark$$

Q2 (a)



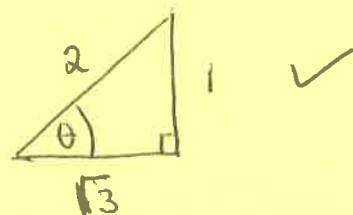
$$\theta = \frac{\pi}{6} \quad \checkmark \checkmark$$

$$(b) \quad \frac{\tan x \cot x}{\cos x} = \frac{1}{\tan x} \cdot \frac{\tan x}{\cos x} \quad \checkmark$$

$$= \frac{1}{\cos x} \quad \checkmark$$

$$= \sec x \quad \checkmark$$

$$(c) \quad \frac{\sqrt{3}}{3} = \frac{3}{3\sqrt{3}} = \frac{1}{\sqrt{3}}$$



$$y = \arctan\left(\frac{\sqrt{3}}{3}\right) \quad \checkmark$$

$$\Rightarrow \tan y = \frac{\sqrt{3}}{3} \quad \checkmark$$

$$\Rightarrow y = \frac{\pi}{6} \quad \checkmark$$

Questio 4

$$a) \int \frac{1+x}{\sqrt{x}} dx = \int x^{-1/2} dx + \int x^{1/2} dx \quad \checkmark \checkmark$$

$$= \frac{x^{1/2}}{1/2} + \frac{x^{3/2}}{3/2} + C \quad \checkmark \checkmark$$

$$= 2\sqrt{x} + \frac{2}{3}x^{3/2} + C \quad \checkmark$$

$$b) \int \sin x \cos x dx$$

$$u = \sin x \quad \checkmark \checkmark$$

$$du = \cos x dx$$

$$= \int \overset{\checkmark}{u} du = \frac{u^2}{2} + C = \frac{\sin^2 x}{2} + C \quad \checkmark$$

QUESTION 3

(1) Find the limits if they exist

$$(a) \lim_{x \rightarrow 0} (2x + \frac{\sin 2x}{x}) = \lim_{x \rightarrow 0} 2x + \lim_{x \rightarrow 0} \frac{\sin 2x}{x} = 0 + 2 = 2. \quad [3]$$

$$(b) \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{(x - 1)(x + 1)}{x - 1} = \lim_{x \rightarrow 1} (x + 1) = 2. \quad [3]$$

(2) $f(x) = \begin{cases} -3ax^2, & x \geq 2 \\ 5 + x & x < 2 \end{cases}$ is continuous on the real line if it is continuous at $x = 2$. That is, if

$$f(2) = -12a = \lim_{x \rightarrow 2^-} (5 + x) = \lim_{x \rightarrow 2^+} (-3ax^2)$$

that is, if $-12a = 7$ or $a = -7/12$ [4]

QUESTION 5

$$(1) y' = \frac{2(x+3) - (2x+1)}{(x+3)^2} = \frac{5}{(x+3)^2} \quad [3]$$

$$(2) y = \cos x - \cos x + x \sin x = x \sin x \quad [3]$$

$$(3) y' = \frac{20}{x}. \quad [2]$$

$$(4) y' = e^x + xe^x = (1 + x)e^x \quad [2]$$

QUESTION 7 Consider the curve $x^2y^2 = 9$.

(1) We have $3^2(-1)^2 = 9$. Thus $(3, -1)$ lies on the curve. [2]

(2) Differentiation both side of $x^2y^2 = 9$, we obtain

$$2xy^2 + 2x^2y \frac{dy}{dx} = 0$$

Solving for $\frac{dy}{dx}$ we have $\frac{dy}{dx} = \frac{y}{x}$. [8]

(3) The slope of the curve at the point $(1, 3)$ is $y'(1, 3) = 3$ [2]

(4) The tangent line to the curve at $(-1, 3)$ is given by $y - 3 = -3(x + 1)$ and the normal line to the curve at $(-1, 3)$ is given by $y - 3 = \frac{1}{3}(x + 1)$. [8]

①

Question 8

$$f(x) = 3x^4 - 4x^3 + 2$$

a) $f'(x) = 12x^3 - 12x^2$ ✓

$$\Rightarrow 12x^3 - 12x^2 = 0$$

$$\Rightarrow 12x^2(x-1) = 0$$

$\Rightarrow x=0$ and $x=1$ are critical numbers. ✓

b)

Interval	$-\infty < x < 0$	$0 < x < 1$	$1 < x < \infty$
Test Value	$x = -1$	$x = \frac{1}{2}$	$x = 2$
Sign of $f'(x)$	-	-	+
Conclusion	decreasing ✓	decreasing ✓	increasing ✓

decreasing $(-\infty, 1)$
 increasing $(1, \infty)$

c) $(1, 1)$ is a relative minimum. ✓

d) $f''(x) = 36x^2 - 24x$ ✓

$$\Rightarrow 36x^2 - 24x = 0$$

$$\Rightarrow 12x(3x-2) = 0$$

$\Rightarrow x=0$ and $x=\frac{2}{3}$ are possible points of inflection

Interval	$-\infty < x < 0$	$0 < x < \frac{2}{3}$	$\frac{2}{3} < x < \infty$ ✓
Test Value	$x = -1$	$x = \frac{1}{2}$	$x = 1$
Sign of $f''(x)$	+	-	+
Conclusion	Concave upward ✓	Concave downward ✓	Concave upward ✓

(2)

Question 8

$$f(0) = 2$$

$$f\left(\frac{2}{3}\right) = \frac{38}{7}$$

$(0, 2)$ and $\left(\frac{2}{3}, \frac{38}{7}\right)$ are points of inflection.

(1)

Question 9

$$a(i) \int \frac{1}{\sqrt{x}(1+\sqrt{x})^2} dx$$

$$u = 1 + x^{\frac{1}{2}} \checkmark$$

$$\frac{du}{dx} = \frac{1}{2} x^{-\frac{1}{2}}$$

$$du = \frac{1}{2} \frac{1}{\sqrt{x}} dx \checkmark$$

$$2 du = \frac{1}{\sqrt{x}} dx$$

$$\int \frac{1}{u^2} \cdot 2 du = 2 \cdot \frac{u^{-1}}{-1} = -2 \frac{1}{u} = \frac{-2}{1+\sqrt{x}} + C$$

$$a(ii) \int e^{\sec 2x} \sec 2x \tan 2x dx$$

$$u = \sec 2x \checkmark$$

$$\frac{du}{dx} = 2 \sec 2x \tan 2x$$

$$\frac{du}{2} = \sec 2x \tan 2x dx \checkmark$$

$$\int e^u \cdot \frac{du}{2} = \frac{1}{2} e^u + C = \frac{1}{2} e^{\sec 2x} + C \checkmark$$

b)

(2)

Question 9

$$b) \int_0^{\pi/3} \frac{\sin x}{(1+\cos x)^2} dx$$

$$u = 1 + \cos x \quad \checkmark$$

$$\frac{du}{dx} = -\sin x$$

$$-du = \sin x \, dx$$

$$\int \frac{-du}{u^2} = - \frac{u^{-1}}{-1} + C = \frac{1}{u} + C$$

$$= \frac{1}{1+\cos x} \bigg|_0^{\pi/3} \quad \checkmark$$

$$= \frac{1}{1+\cos \pi/3} - \frac{1}{1+\cos 0} \quad \checkmark$$

$$= \frac{1}{1+\frac{1}{2}} - \frac{1}{2}$$

$$= \frac{1}{3/2} - \frac{1}{2}$$

$$= \frac{2}{3} - \frac{1}{2} = \frac{4-3}{6} = \frac{1}{6} \quad \checkmark$$

Question 6

$$f(x) = 2x^3 - 6x \quad [0, 3]$$

$$\begin{aligned} f'(x) &= 6x^2 - 6 \checkmark \Rightarrow 6x^2 - 6 = 0 \\ &\Rightarrow 6(x^2 - 1) = 0 \\ &\Rightarrow 6(x-1)(x+1) = 0 \checkmark \\ &\Rightarrow \underset{\checkmark}{x=1} \text{ and } \underset{\checkmark}{x=-1} \text{ are critical numbers} \\ &\hspace{15em} \text{in the interval } [0, 3] \end{aligned}$$

$$f(0) = 0 \checkmark$$

$$f(-1) = 4 \checkmark$$

$$f(1) = -4 \checkmark$$

$$f(3) = 36 \checkmark \quad \text{Maximum}$$

$$\text{Minimum is } -4 \checkmark$$

$$\text{Maximum is } 36 \checkmark$$

$$(a) \ y^2 = -6x$$

$$= 4\left(-\frac{6}{4}\right)x \quad \checkmark$$

$$V(0,0), F(0, -\frac{3}{2}) \quad \checkmark \checkmark$$

$$\text{Directrix at } x = \frac{3}{2} \quad \checkmark$$

$$(b) \ \sin 2x \sin x = 2 \cos x$$

$$\Rightarrow 2 \cos x \sin x \sin x = 2 \cos x \quad \checkmark$$

$$\Rightarrow \cos x \cdot \sin^2 x - \cos x = 0 \quad \checkmark$$

$$\Rightarrow \cos x (\sin^2 x - 1) = 0 \quad \checkmark$$

$$\Rightarrow \cos x = 0 \text{ or } \sin x = \pm 1 \quad \checkmark \checkmark$$

$$\cos x = 0 \Rightarrow x = \frac{\pi}{2}, \frac{3\pi}{2} \quad \checkmark \checkmark$$

$$\sin x = 1 \Rightarrow x = \frac{\pi}{2}, \frac{3\pi}{2} \quad \checkmark \checkmark$$

$$\therefore x = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$(c) \ x^2 - 9y^2 + 36y - 72 = 0$$

$$\Rightarrow x^2 - 9(y^2 - 4y) - 72 = 0$$

$$\Rightarrow x^2 - 9(y-2)^2 - 9 \cdot -4 - 72 = 0 \quad \checkmark \checkmark$$

$$\Rightarrow x^2 - 9(y-2)^2 - 36 = 0 \quad \checkmark$$

$$\Rightarrow x^2 - 9(y-2)^2 = 36$$

$$\Rightarrow \frac{x^2}{36} - \frac{(y-2)^2}{4} = 1 \quad \checkmark$$

$$C(0,2), V_1(6,2) \quad \checkmark \checkmark$$

$$V_2(0,-4) \quad \checkmark$$

UNIVERSITY OF BOTSWANA
DEPARTMENT OF MATHEMATICS
2017/18 – SEMESTER II
EXAMINATIONS

FRONT PAGE

COURSE NUMBER MAT122 **DURATION** 2 hrs **DATE** May 2018
TITLE OF PAPER INTRODUCTORY MATHEMATICS II
SUBJECT MATHEMATICS **TITLE OF EXAMINATION** BSc I

MORNING/AFTERNOON

INSTRUCTIONS:

- ANSWER ALL QUESTIONS IN SECTION A AND ANY TWO FROM SECTION B.
- ALL MARKS ARE INDICATED IN [].

NUMBER OF PAGES: 5

**DO NOT OPEN THIS PAGE UNTIL YOU HAVE BEEN TOLD
TO DO SO BY THE SUPERVISOR.**

Section A

Answer All Questions: Each Question carries 10 Marks

Question 1

Evaluate the following limits:

(a) $\lim_{x \rightarrow 0} \frac{x}{\sqrt{x+1}}$ [2]

(b) $\lim_{x \rightarrow 5} \frac{x-5}{x^2-25}$ [4]

(c) $\lim_{t \rightarrow 0} \frac{\sin 3t}{\sin 2t}$ [4]

Question 2

(a) Find $\frac{dy}{dx}$ given that $x^2 + y^2 = 1$. [3]

(b) Find $\frac{dy}{dx}$ for the parametric equations, $x = 1 - \sin \theta$, $y = 1 - \cos \theta$. [3]

(c) Find the equation of the parabola with the focus at $(-4, 1)$ and the directrix $x = 0$. [4]

Question 3

(a) Differentiate the following functions:

(i) $y = \sqrt{x} \sin x$. [3]

(ii) $y = \frac{x^2 - 5x}{3 + x^7}$. [3]

(b) Find $\frac{d^2y}{dx^2}$, given $y = 2xe^{-x}$. [4]

Question 4

(a) Find the horizontal asymptote of the function $f(x) = \frac{x+2}{2x-2}$. [3]

(b) Determine the absolute maximum and minimum values of

$$f(x) = 3x^3 - 9x$$

on the interval $[0,3]$. [7]

Question 5

Evaluate the following integrals

(a) $\int (6x^2 + \frac{3}{x} + 5 \sin x) dx$. [3]

(b) $\int \frac{2x^2}{x^3 + 3} dx$. [3]

(c) $\int_1^2 (3x^2 + 5) dx$. [4]

Question 6

(a) Show that $f(x) = |x - 6|$ is not differentiable at $x = 6$. [5]

(b) Use logarithmic differentiation to find $\frac{dy}{dx}$ for $y = x\sqrt{x^2 + 1}$, $x > 0$. [5]

Section B
Answer TWO Questions OUT of FOUR:
Each Question carries 20 Marks

Question 7

- (a) Integrate the following :

(i) $\int_0^{\pi/2} x \cos x dx.$ [6]

(ii) $\int \frac{1}{(x+1)(x-1)} dx.$ [6]

- (b) Find the area of the region bounded by the curves:
 $y = x^2 - 7x + 7$ and $y = 1 - 2x.$ [8]

Question 8

- (a) Describe the conic section with equation $9x^2 - 4y^2 + 54x + 8y + 41 = 0$
and determine all of its key features. [11]

- (b) Show that the function

$$f(x) = \begin{cases} 2x^2 + 4, & x \leq 0 \\ (x-2)^2, & x > 0 \end{cases}$$

is continuous at $x = 0.$ [6]

- (c) Evaluate $\lim_{x \rightarrow \infty} \frac{3x^3 - 4x^2 + 1}{x^2 + 3}.$ [3]

Question 9

- (a) Find the equation of the tangent line to the curve $y = \ln x$ at the point $(1, 0)$. [5]
- (b) Find $\frac{dy}{dx}$ by implicitly differentiating $e^{xy} + 4x = y + x^4$. [5]
- (c) Find the particular solution of $F'(x) = e^x + 2$ with the initial condition $F(0) = 2$. [5]
- (d) Find $\frac{dz}{d\theta}$ if $z = 5 \sin 2\theta \cos^2 \theta$. [5]

Question 10

Given that $f(x) = 3x^2 - x^3$

- (a) Find the x -intercepts and the y -intercepts. [3]
- (b) Find and classify the critical points. [8]
- (c) Find the intervals on which the graph of f is concave up or concave down. [4]
- (d) Hence sketch the graph of the function f . [5]

DEPARTMENT OF MATHEMATICS

Course Moderation Form

ACADEMIC YEAR: 2017/2018 **SEMESTER:** I

COUSE CODE & TITLE: MAT122 INTRODUCTORY MATHEMATICS II

LECTURER: Dr M. MOTHEBE, Dr R. TSHEKAMETSE, Mr S. LEKOKO

MODERATORS: _____

Lecture Notes (reflect standards/course outline) _____

Assignments (general comments) _____

Assignment/Test (general comments): _____

Test I (seen/signed/dated): _____

Test II (seen/signed/dated): _____

C.A. based on: _____

Final Exam: _____

Overall agreement for the course between Lecturer and Moderator:

SIGNATURES

Lecturer: _____ **Date:** _____

Moderator I: R. TSHEKAMETSE **Date:** _____

Moderator II: M. MOTHEBE **Date:** _____

UNIVERSITY OF BOTSWANA
DEPARTMENT OF MATHEMATICS
2017/2018 -SUPPLEMENTARY EXAMINATIONS

FRONT PAGE

COURSE NUMBER: MAT122 **DURATION:** 2 hrs **DATE:** Jan 2017
TITLE OF PAPER: INTRODUCTORY MATHEMATICS II
SUBJECT: MATHEMATICS **TITLE OF EXAMINATION:** BSc I

MORNING/AFTERNOON

INSTRUCTIONS:

- ANSWER ALL QUESTIONS IN SECTION A AND ANY TWO QUESTIONS FROM SECTION B.
- ALL MARKS ARE INDICATED IN [].

NUMBER OF PAGES: 6

**DO NOT OPEN THIS PAGE UNTIL YOU HAVE BEEN TOLD TO DO
SO BY THE SUPERVISOR.**

SECTION A

Answer All Questions: Each Question carries 10 Marks

Question 1

- (a) Let $(-3, 4)$ be the point on the terminal side of an angle θ in standard position. Find $\sin \theta$. [4]

- (b) Find all solutions of

$$2 \sin \theta + \sqrt{2} = -\sin \theta,$$

in the interval $[0, 360^\circ]$

[6]

Question 2

- (a) Find the limit

$$\lim_{x \rightarrow 4} \frac{x - 4}{\sqrt{x + 5} - 3}.$$

[6]

- (b) Find the derivative

$$y = (\sin x + e^{2x} + \ln x)^{\frac{1}{3}}.$$

[4]

Question 3

(a) Find $\frac{dy}{dx}$ and the slope at the given point

$$x = e^t, \quad y = 2e^t + t, \quad t = \ln 2.$$

[5]

(b) Discuss the concavity of the graph of the function

$$f(x) = x^3 - 3x - 1.$$

[5]

Question 4

(a) Evaluate the integral

$$\int \frac{1}{\sqrt{x}} + \sec^2 x - e^{2x} dx.$$

[4]

(b) Show that $\int_0^{\frac{\pi}{2}} \frac{\cos x}{1 + \sin x} dx = \ln 2.$

[6]

Question 5

- (a) Use integration by parts to evaluate

$$\int x \ln x \, dx.$$

[5]

- (b) Find the area of the region bounded by the curves
 $y = x^2 - 2$ and $y = x$.

[5]

Question 6

- (a) Evaluate the integral $\int \frac{x+1}{x^2+x-2} \, dx$ by partial fractions. [6]

- (b) Evaluate the integral $\int \frac{x^2+2x-1}{x^2-1} \, dx$. [4]

SECTION B

Answer TWO Questions OUT of FOUR:
Each Question carries 20 Marks

Question 7

- (a) Find all solutions of

$$\cos^3 x = \cos x$$

in the interval $[0, 2\pi)$.

[7]

- (b) Let $x^3 + \sin(xy) - y^2 = 10$. Find $\frac{dy}{dx}$ implicitly.

[7]

- (c) Discuss if the function $f(x)$ is continuous at $x = 1$ given that

[6]

$$f(x) = \begin{cases} x^2 + 2, & x \geq 1, \\ 4x - 1, & x < 1, \end{cases}$$

Question 8

Given

$$f(x) = \frac{1}{6} \left(\frac{x^4}{2} - x^3 \right).$$

- (a) Find the intervals on which f is increasing and where it is decreasing.

[6]

- (b) Identify where the extrema of f occur.

[3]

- (c) Find the intervals on which f is concave up and where it is concave down.

[5]

- (d) Sketch the graph showing the local maximum and minimum points, points of inflection and y -intercept.

[6]

Question 9

(a) By writing $\tan^4 x = \tan^2 x(\sec^2 x - 1)$ show that

$$\int_0^{\frac{\pi}{4}} \tan^4 x \, dx = \frac{\pi}{4} - \frac{2}{3}.$$

[8]

(b) Evaluate $\int x^2 e^x \, dx$ by parts.

[6]

(c) Show that $\int_0^2 \frac{x}{\sqrt{2x^2 + 1}} \, dx = 1.$

[7]

Question 10

(a) Use the substitution $x = 3 \tan \theta$ to find

$$\int_0^3 \frac{x^3}{\sqrt{x^2 + 9}} \, dx.$$

[12]

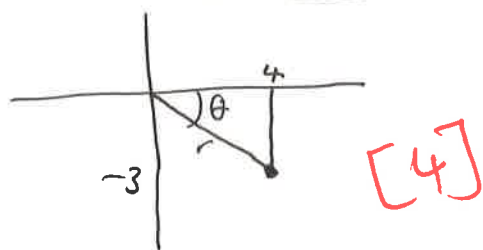
(b) By the use of natural logarithm function find $\frac{dy}{dx}$ of

[8]

$$y = \frac{x^2 \sqrt{2x - 3}}{(x + 1)^2}.$$

QUESTION 1

(a)



$$r = \sqrt{(4)^2 + (-3)^2} = 5$$

$$\sin \theta = \frac{y}{r} = \frac{-3}{5}$$

(b) $2\sin \theta + \sqrt{2} = -\sin \theta$

$$\sin \theta = -\frac{\sqrt{2}}{3}$$

$$\theta = \sin^{-1}\left(-\frac{\sqrt{2}}{3}\right)$$

$$= -28.1^\circ$$

$\sin \theta < 0$ on the 3rd and 4th quadrant then

$$\theta = 360^\circ - 28.1^\circ$$

$$= 331.9^\circ$$

and

$$\theta = 180^\circ + 28.1^\circ$$

$$= 208.1^\circ$$

QUESTION 2

(a) $\lim_{x \rightarrow 4} \frac{x-4}{\sqrt{x+5}-3}$

$$= \lim_{x \rightarrow 4} \frac{x-4}{\sqrt{x+5}-3} \cdot \frac{\sqrt{x+5}+3}{\sqrt{x+5}+3}$$

$$= \lim_{x \rightarrow 4} \frac{(x-4)(\sqrt{x+5}+3)}{(x+5)-9}$$

$$= \lim_{x \rightarrow 4} \frac{(x-4)(\sqrt{x+5}+3)}{x-4}$$

$$= \lim_{x \rightarrow 4} (\sqrt{x+5}+3)$$

$$= \sqrt{4+5}+3$$

$$= 6$$

(b) $y = (\sin x + e^{2x} + \ln x)^{\frac{1}{3}}$

$$\frac{dy}{dx} = \frac{1}{3} (\sin x + e^{2x} + \ln x)^{-\frac{2}{3}} \frac{d}{dx} (\sin x + e^{2x} + \ln x)$$

$$= \frac{1}{3} (\sin x + e^{2x} + \ln x)^{-\frac{2}{3}} (\cos x + 2e^{2x} + \frac{1}{x})$$

[4]

QUESTION 3

(a) $x = e^t$, $y = e^t + t$; $t = \ln 2$

$\frac{dx}{dt} = e^t$ and $\frac{dy}{dt} = e^t + 1$

$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{e^t + 1}{e^t} = 1 + e^{-t}$

$\left. \frac{dy}{dx} \right|_{t=\ln 2} = 1 + e^{-\ln 2}$

$[5] = 1 + \frac{1}{2} = \frac{3}{2}$

(b) $y = x^3 - 3x - 1$

$y' = 3x^2 - 3$

$y'' = 6x$

$y'' = 0 \Rightarrow x = 0$

$(-\infty, 0)$	$(0, \infty)$
-100	100
y'' -	+

The $y =$ is concave up on $(0, \infty)$
and is concave down on $(-\infty, 0)$

[5]

QUESTION 4

(a) $\int \left(\frac{1}{\sqrt{x}} + \sec^2 x - e^{2x} \right) dx$

$= \int \left(x^{-\frac{1}{2}} + \sec^2 x - e^{2x} \right) dx$ [4]

$= 2x^{\frac{1}{2}} + \tan x - \frac{e^{2x}}{2} + C$

(b) $\int_0^{\frac{\pi}{2}} \frac{\cos x}{1 + \sin x} dx = I$

Let $u = 1 + \sin x \Rightarrow du = \cos x dx$

$\begin{array}{c|cc} x & 0 & \frac{\pi}{2} \\ \hline u & 1 & 2 \end{array}$

$\therefore I = \int_1^2 \frac{1}{u} du$ [6]

$= \ln u \Big|_1^2$

$= \ln 2 - \ln 1$

$= \ln 2$

OR

$\int \frac{\cos x}{1 + \sin x} dx = I$

Let $u = 1 + \sin x$

$du = \cos x dx$

$\therefore I = \int \frac{1}{u} du$

$= \ln u + C$

$= \ln |1 + \sin x| + C$

$I \Big|_0^{\frac{\pi}{2}} = \ln |1 + \sin x| \Big|_0^{\frac{\pi}{2}}$

$= \ln (1 + \sin \frac{\pi}{2}) - \ln (1 + \sin 0)$

$= \ln 2$

QUESTION 5

$$1) \int x \ln x dx = uv - \int v du$$

$$\text{let } u = \ln x \quad dv = x dx$$

$$du = \frac{1}{x} dx \quad v = \frac{x^2}{2}$$

$$\therefore I = \frac{x^2}{2} \ln x - \int \frac{x^2}{2} \cdot \frac{1}{x} dx$$

$$[5] = \frac{x^2}{2} \ln x - \frac{1}{2} \int x dx$$

$$= \frac{x^2}{2} \ln x - \frac{1}{2} \cdot \frac{x^2}{2} + C$$

$$= \frac{x^2}{2} \ln x - \frac{1}{4} x^2 + C$$

(b) Intersection

$$x^2 - 2 = x \Rightarrow x^2 - x - 2 = 0$$

$$\Rightarrow (x+1)(x-2) = 0$$

$$\Rightarrow x = -1, 2$$

$$x > x^2 - 2 \quad \forall x \in (-1, 2)$$

$$\therefore \text{Area} = \int_{-1}^2 x - (x^2 - 2) dx$$

$$= \int_{-1}^2 x - x^2 + 2 dx$$

$$[5] = \left. \frac{x^2}{2} - \frac{x^3}{3} + 2x \right|_{-1}^2$$

$$= \left(2 - \frac{8}{3} + 4 \right) - \left(\frac{1}{2} + \frac{1}{3} - 2 \right)$$

$$= \underline{\underline{\frac{9}{2}}}$$

QUESTION 6

$$(a) \frac{x+1}{(x+2)(x-1)} = \frac{A}{x+2} + \frac{B}{x-1}$$

$$x+1 = A(x-1) + B(x+2)$$

$$= (A+B)x - A + 2B$$

$$A+B = 1 \quad \text{--- (1)}$$

$$-A+2B = 1 \quad \text{--- (2)}$$

$$(1) + (2) \Rightarrow 3B = 2 \Rightarrow B = \frac{2}{3}$$

$$(1) \Rightarrow A = 1 - B = \frac{1}{3}$$

$$\therefore \int \frac{x+1}{(x+2)(x-1)} dx = \frac{1}{3} \int \frac{1}{x+2} dx + \frac{2}{3} \int \frac{1}{x-1} dx$$

$$[6] = \frac{1}{3} \ln|x+2| + \frac{2}{3} \ln|x-1| + C$$

$$(b) \int \frac{x^2+2x-1}{x^2-1} dx = \int \frac{x^2-1+2x}{x^2-1} dx$$

$$= \int \left(1 + \frac{2x}{x^2-1} \right) dx$$

$$= x + \ln|x^2-1| + C$$

$$\text{let } u = x^2 - 1$$

$$du = 2x dx$$

$$\therefore I = x + \int \frac{1}{u} du$$

$$= x + \ln u + C$$

$$= x + \ln(x^2-1) + C$$

[4]

QUESTION 7

(a) $\cos^3 x = \cos x$

$$\cos^3 x - \cos x = 0 \quad \checkmark \quad [7]$$

$$\cos x (\cos^2 x - 1) = 0$$

$$\cos x = 0 \quad \checkmark \quad \text{or} \quad \cos x = \pm 1 \quad \checkmark$$

$$x = \cos^{-1}(0) \quad \cos x = 1 \Rightarrow x = 0^\circ \quad \checkmark$$

$$= 90^\circ, 270^\circ \quad \checkmark \quad \text{or} \quad 360^\circ \quad \checkmark$$

$$\cos x = -1 \Rightarrow x = 180^\circ \quad \checkmark$$

$$x = 0^\circ, 90^\circ, 180^\circ, 270^\circ, 360^\circ$$

(b) $x^3 + \sin(xy) - y^3 = 10$

differentiate both sides wrt x

$$3x^2 + \cos xy \frac{d}{dx}(xy) - 3y^2 \frac{dy}{dx} = 0 \quad \checkmark$$

$$3x^2 + \cos xy \left[y + x \frac{dy}{dx} \right] - 3y^2 \frac{dy}{dx} = 0$$

$$y \cos xy + x \cos xy \frac{dy}{dx} - 3y^2 \frac{dy}{dx} = -3x^2$$

$$(x \cos xy - 3y^2) \frac{dy}{dx} = -3x^2 - y \cos xy \quad \checkmark$$

$$\therefore \frac{dy}{dx} = \frac{-3x^2 - y \cos xy}{x \cos xy - 3y^2} \quad \checkmark$$

[7]

$$(c) f(x) = \begin{cases} x^2 + 2, & x \geq 1 \\ 4x - 1, & x < 1 \end{cases}$$

$$f(1) = 1^2 + 2 = 3 \quad \checkmark$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (4x - 1)$$

$$= 4 - 1 \quad \checkmark$$

$$= 3 \quad \checkmark$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (x^2 + 2)$$

$$= 1 + 2 \quad \checkmark$$

$$= 3 \quad \checkmark$$

$$\therefore \lim_{x \rightarrow 1} f(x) = 3 \quad \checkmark$$

$$\text{As } f(1) = \lim_{x \rightarrow 1} f(x) = 3 \quad \checkmark \text{ then}$$

the function is continuous at $x = 1$.

[6]

QUESTION 8

$$f(x) = \frac{1}{6} \left(\frac{x^4}{2} - x^3 \right)$$

$$(a) f'(x) = \frac{1}{6} \left(\frac{4x^3}{2} - 3x^2 \right) \\ = \frac{1}{6} (2x^3 - 3x^2)$$

$$f' = 0 \Rightarrow 2x^3 - 3x^2 = 0$$

$$\Rightarrow x^2(2x - 3) = 0$$

$$\Rightarrow x = 0, \frac{3}{2}$$

$$(-\infty, 0) \quad (0, \frac{3}{2}) \quad (\frac{3}{2}, \infty)$$

-100

1

100

+

+

+

-

-

-

+

+

+

-

-

-

+

+

+

-

-

-

+

+

+

-

-

-

+

+

+

-

-

-

+

+

+

-

-

-

+

+

+

-

-

The $f(x)$ is increasing on $(\frac{3}{2}, \infty)$ and is decreasing on $(-\infty, \frac{3}{2})$.

$$(b) \begin{array}{c|c|c} x & 0 & \frac{3}{2} \\ \hline y & 0 & -\frac{9}{16} \end{array}$$

$\therefore (\frac{3}{2}, -\frac{9}{16})$ is a local min.

$(0, 0)$ is ^{neither} a local min nor a local max

$$(c) f''(x) = \frac{1}{6} (6x^2 - 6x)$$

$$f'' = 0 \Rightarrow x^2 - x = 0$$

$$\Rightarrow x(x-1) = 0$$

$$\Rightarrow x = 0, 1$$

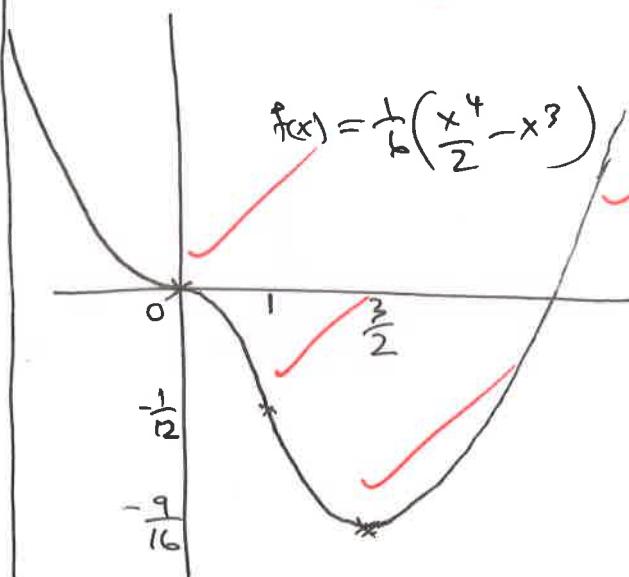
	$(-\infty, 0)$ -100	$(0, 1)$ 0.5	$(1, \infty)$ 100
x	-	+	+
x-1	-	-	+
f'(x)	+	-	+

$\therefore$ the $f(x)$ is concave up on $(-\infty, 0)$ and $(1, \infty)$ and is concave down on $(0, 1)$.

(d) pt of inflection

x	0	1
y	0	$-\frac{1}{12}$

Sketch



QUESTIONS 9

$$(a) \int_0^{\frac{\pi}{4}} \tan^4 x \, dx$$

$$= \int_0^{\frac{\pi}{4}} \tan^2 x (\sec^2 x - 1) \, dx$$

$$= \int_0^{\frac{\pi}{4}} (\tan^2 x \sec^2 x - \tan^2 x) \, dx$$

$$= \int_0^{\frac{\pi}{4}} \tan^2 x \sec^2 x \, dx - \int_0^{\frac{\pi}{4}} \tan^2 x \, dx$$

$$= \int_0^{\frac{\pi}{4}} \tan^2 x \sec^2 x \, dx - \int_0^{\frac{\pi}{4}} (\sec^2 x - 1) \, dx$$

$$\text{let } u = \tan x$$

$$du = \sec^2 x \, dx$$

$$\therefore I = \int u^2 \, du - (\tan x - x)$$

$$= \frac{u^3}{3} - (\tan x - x)$$

$$= \frac{\tan^3 x}{3} - \tan x + x \Big|_0^{\frac{\pi}{4}}$$

$$= \frac{\tan^3 \frac{\pi}{4}}{3} - \tan \frac{\pi}{4} + \frac{\pi}{4}$$

$$- \left(\frac{\tan^3 0}{3} - \tan 0 + 0 \right)$$

$$= \frac{1}{3} - 1 + \frac{\pi}{4}$$

$$= \frac{\pi}{4} - \frac{2}{3}$$

$$(b) \int x^2 e^{2x} \, dx = uv - \int v \, du$$

$$u = x^2 \quad dv = e^{2x} \, dx$$

$$du = 2x \, dx \quad v = e^{2x}$$

$$\therefore I = x^2 e^{2x} - 2 \int x e^{2x} \, dx$$

$$u = x \quad dv = e^{2x} \, dx$$

$$du = dx \quad v = e^{2x}$$

$$\therefore I = x^2 e^{2x} - 2(uv - \int v \, du)$$

$$= x^2 e^{2x} - 2(xe^{2x} - \int e^{2x} \, dx)$$

$$= x^2 e^{2x} - 2xe^{2x} + 2e^{2x} + C$$

$$= (x^2 - 2x + 2)e^{2x} + C$$

$$(c) \int_0^2 \frac{x}{\sqrt{2x^2+1}} \, dx$$

$$\text{let } u = 2x^2 + 1 \Rightarrow du = 4x \, dx$$

$$\Rightarrow \frac{1}{4} du = x \, dx$$

x	0	2
u	1	9

$$\therefore I = \int_1^9 u^{-\frac{1}{2}} \cdot \frac{1}{4} \, du$$

$$= \frac{1}{4} \frac{u^{\frac{1}{2}}}{\frac{1}{2}} \Big|_1^9$$

$$= \frac{2}{4} (9^{\frac{1}{2}} - 1^{\frac{1}{2}})$$

$$= \frac{1}{2} (3 - 1)$$

$$= 1$$

QUESTION 10

$$(a) \int_0^3 \frac{x^3}{\sqrt{x^2+9}} dx = I$$

$$x = 3 \tan \theta \Rightarrow dx = 3 \sec^2 \theta d\theta$$

$$\Rightarrow \theta = \tan^{-1}\left(\frac{x}{3}\right)$$

x	0	3
θ	0	$\frac{\pi}{4}$

$$\sqrt{x^2+9} = \sqrt{(3 \tan \theta)^2 + 9}$$

$$= 3 \sec \theta$$

$$x^3 = 27 \tan^3 \theta$$

$$\therefore I = \int_0^{\frac{\pi}{4}} \frac{27 \tan^3 \theta \cdot 3 \sec^2 \theta d\theta}{3 \sec \theta}$$

$$= 27 \int_0^{\frac{\pi}{4}} \tan^3 \theta \sec \theta d\theta$$

$$= 27 \int_0^{\frac{\pi}{4}} \tan^2 \theta \sec \theta \tan \theta d\theta$$

$$= 27 \int_0^{\frac{\pi}{4}} (\sec^2 \theta - 1) \sec \theta \tan \theta d\theta$$

$$u = \sec \theta \Rightarrow du = \sec \theta \tan \theta d\theta$$

θ	0	$\frac{\pi}{4}$
u	1	$\frac{2}{\sqrt{2}}$

$$\therefore I = 27 \int_1^{\frac{2}{\sqrt{2}}} (u^2 - 1) du$$

$$= 27 \left(\frac{u^3}{3} - u \right) \Big|_1^{\frac{2}{\sqrt{2}}}$$

$$= 27 \left[\frac{4}{3\sqrt{2}} - \frac{2}{\sqrt{2}} - \left(\frac{1}{3} - 1 \right) \right]$$

$$= 18 \left(\frac{\sqrt{2}-1}{\sqrt{2}} \right)$$

$$(b) y = \frac{x^2 \sqrt{2x-3}}{(x-1)^2}$$

$$\ln y = \ln \left(\frac{x^2 \sqrt{2x-3}}{(x-1)^2} \right)$$

$$= 2 \ln x + \frac{1}{2} \ln (2x-3) - 2 \ln (x-1)$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{2 \frac{d}{dx}(x)}{x} + \frac{1}{2} \frac{\frac{d}{dx}(2x-3)}{2x-3} - 2 \frac{\frac{d}{dx}(x-1)}{x-1}$$

$$= \frac{2}{x} + \frac{1}{2} \left(\frac{2}{2x-3} \right) - \frac{2}{x-1}$$

$$\therefore \frac{dy}{dx} = y \left(\frac{2}{x} + \frac{1}{2x-3} - \frac{2}{x-1} \right)$$

$$= \frac{x^2 \sqrt{2x-3}}{(x-1)^2} \left(\frac{2}{x} + \frac{1}{2x-3} - \frac{2}{x-1} \right)$$

[8]

UNIVERSITY OF BOTSWANA
DEPARTMENT OF MATHEMATICS
2016/2017 – EXAMINATIONS

FRONT PAGE

COURSE NUMBER: MAT122 **DURATION:** 2 hrs **DATE:** DEC 2016

TITLE OF PAPER: INTRODUCTORY MATHEMATICS II

SUBJECT: MATHEMATICS **TITLE OF EXAMINATION:** BSc I

MORNING/AFTERNOON

INSTRUCTIONS:

- ANSWER ALL QUESTIONS IN SECTION A AND ANY TWO QUESTIONS FROM SECTION B.
- ALL MARKS ARE INDICATED IN [].

NUMBER OF PAGES: 6

**DO NOT OPEN THIS PAGE UNTIL YOU HAVE BEEN TOLD TO DO
SO BY THE SUPERVISOR.**

SECTION A

Answer All Questions: Each Question carries 10 Marks

Question 1

(a) Evaluate

$$\lim_{x \rightarrow 0} \frac{\tan 7x}{\sin 3x}.$$

[5]

(b) Find the constant k that makes the function

$$f(x) = \begin{cases} x^2 - kx + 6, & x < 3, \\ -x^2 - kx - 2, & x \geq 3. \end{cases}$$

continuous at $x = 3$.

[5]

Question 2

(a) Let $f(x) = 2x^2 + 1$. Use the limit process to find the derivative of $f(x)$.

[5]

(b) Find the derivative of the function

$$f(x) = \frac{1 - \cos x}{\sin x},$$

at $x = \frac{\pi}{3}$.

[5]

Question 3

Given that $x + xy - 2x^3 = 2$,

- (a) Find $\frac{dy}{dx}$ by implicit differentiation. [3]
- (b) Solve the equation for y as a function of x , then find $\frac{dy}{dx}$ from the equation. [3]
- (c) Confirm that the two results are consistent by expressing the derivative in part (a) as a function of x alone. [4]

Question 4

Evaluate the following integrals:

- (a) $\int \frac{x^2 - 2x}{\sqrt{x}} dx$, [4]
- (b) $\int_1^2 xe^x dx$. [6]

Question 5

(a) Evaluate $\int (\cos x)e^{\sin x} dx$. [4]

(b) Find the area bounded by the curves $y = x^2 - 2$ and $y = -x$. [6]

Question 6

(a) Evaluate $\int \frac{dx}{x^2 + x}$. [6]

(b) Find $F'(x)$ given that

$$F(x) = \int_2^{x^2} (\cos t) dt.$$

[4]

SECTION B

Answer TWO Questions OUT of FOUR:

Each Question carries 20 Marks

Question 7

Given

$$f(x) = x^3 - 6x^2 + 9x.$$

- (a) Identify where the extrema of f occur. [7]
- (b) Find the intervals on which f is increasing and where it is decreasing. [4]
- (c) Find the intervals on which f is concave up and where it is concave down. [5]
- (d) Sketch the graph showing the local maximum and minimum points, points of inflection and y -intercept [4]

Question 8

- (a) Find the equation of the tangent line to the curve

$$x = 2t + 4, \quad y = 8t^2 - 2t + 4$$

at $t = 1$ without eliminating the parameter. [4]

- (b) Find the equation of the tangent line in part (a) by eliminating the parameter. [5]
- (c) Find the value of the constant A so that $y = A \sin 3x$ satisfies the equation

$$\frac{d^2y}{dx^2} + 2y = 4 \sin 3x.$$

[5]

- (d) Find the derivative of

$$y = \ln \left(\frac{\sqrt{x} \cos x}{(1 + x^2)^3} \right).$$

[6]

Question 9

(a) Use integration by substitution to show that

$$\int_0^2 2x^2 \sqrt{x^3 + 1} \, dx = \frac{104}{9}.$$

[7]

(b) Integrate $\int x^2 \ln x \, dx$ by parts.

[5]

(c) Show that the area bounded by the parabola $y = x^2 - 6x + 5$ and the line $y = x - 5$ is $\frac{9}{2}$.

[8]

Question 10

(a) Given that

$$\frac{1}{x^2 - a^2} = \frac{A}{x - a} + \frac{B}{x + a}$$

find A and B in terms of a , and evaluate:

$$\int \frac{dx}{x^2 - a^2}.$$

Deduce that $\int_0^1 \frac{dx}{x^2 - 4} = -\frac{\ln 3}{4}.$

[10]

(b) Find $\int \sqrt{4 - x^2} \, dx$, and show that

$$\int_0^2 \sqrt{4 - x^2} \, dx = \pi.$$

[10]

122

MAT 122

SOL^NS

QUESTION 1

$$(a) \lim_{x \rightarrow 0} \frac{\tan 7x}{\sin 3x} = \lim_{x \rightarrow 0} \frac{\sin 7x}{\cos 7x} \cdot \frac{1}{\sin 3x}$$

NB: $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

NB

$$\frac{\tan 7x}{\sin 3x} = \frac{\sin 7x}{\cos 7x} \cdot \frac{1}{\sin 3x} \checkmark$$

$$\frac{\tan 7x}{\sin 3x} = \frac{1}{\cos 7x} \cdot \frac{\sin 7x}{7x} \cdot 7x \cdot \frac{3x}{\sin 3x} \cdot \frac{1}{3x} \checkmark$$

as $x \rightarrow 0$, $3x \rightarrow 0$ & $7x \rightarrow 0$

So that

$$\lim_{x \rightarrow 0} \frac{\tan 7x}{\sin 3x} = \lim_{x \rightarrow 0} \frac{1}{\cos 7x} \cdot \frac{\sin 7x}{7x} \cdot \frac{3x}{\sin 3x} \cdot \frac{7x}{3x} \checkmark$$

$$= 1 \cdot 1 \cdot 1 \cdot \frac{7}{3} \checkmark \quad [5]$$

$$= \frac{7}{3} \checkmark$$

$$(b) f(x) = \begin{cases} x^2 - kx + 6, & x < 3 \\ -x^2 - kx - 2, & x \geq 3 \end{cases}$$

For $f(x)$ to be cont. at $x=3$ then

$$x^2 - k_1x + 6 = -x^2 - k_2x - 2 \checkmark$$

$$\Rightarrow (3)^2 - k_1(3) + 6 = -(3)^2 - k_2(3) - 2 \checkmark \quad [5]$$

$$\Rightarrow 9 - 3k_1 + 6 = -9 - 3k_2 - 2 \checkmark$$

$$\Rightarrow 15 - 3k_1 = -11 - 3k_2 \checkmark$$

$$\Rightarrow 24 = 6k \checkmark \quad \text{because } k_1 = k_2$$

$$\Rightarrow k = 4 \checkmark$$

Q

$$(a) f(x) = 2x^2 + 1$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{[(2(x+h))^2 + 1] - [2x^2 + 1]}{h}$$

$$= \lim_{h \rightarrow 0} \frac{[2(x^2 + 2xh + h^2) + 1] - [2x^2 + 1]}{h}$$

$$= \lim_{h \rightarrow 0} \frac{4xh + 2h^2}{h}$$

$$= \lim_{h \rightarrow 0} (4x + 2h)$$

$$= 4x$$

[5]

$$(b) f(x) = \frac{1 - \cos x}{\sin x} = \frac{t}{b}$$

$$f'(x) = \frac{bt' - tb'}{b^2}$$

$$= \frac{(\sin x)(\sin x) - (1 - \cos x)(\cos x)}{\sin^2 x}$$

$$= \frac{\sin^2 x - \cos x + \cos^2 x}{\sin^2 x}$$

$$= \frac{1 - \cos x}{\sin^2 x} = \frac{1}{\sin^2 x} - \frac{\cos x}{\sin x} \cdot \frac{1}{\sin x}$$

$$= \operatorname{cosec}^2 x - \cot x \cdot \operatorname{csc} x$$

$$= \operatorname{csc}^2 x - \cot x \operatorname{csc} x$$

$$f'\left(\frac{\pi}{3}\right) = \frac{1 - \cos\left(\frac{\pi}{3}\right)}{\left(\sin \frac{\pi}{3}\right)^2} = \frac{1 - \left(\frac{1}{2}\right)}{\left(\frac{\sqrt{3}}{2}\right)^2} = \frac{1}{2} \div \frac{3}{4}$$

$$= \frac{1}{2} \cdot \frac{4}{3} = \frac{2}{3}$$

[5]

QUESTION 3

$$x + xy - 2x^3 = 2$$

$$(a) \quad x + xy - 2x^3 = 2$$

$$\Rightarrow \frac{d}{dx}(x) + x \frac{dy}{dx} + y \frac{dx}{dx} - 2 \cdot 3x^2 \frac{dx}{dx} = 0 \quad \checkmark$$

$$\Rightarrow 1 + x \frac{dy}{dx} + y - 6x^2 = 0 \quad \checkmark$$

$$\Rightarrow x \frac{dy}{dx} = 6x^2 - y - 1$$

$$\Rightarrow \frac{dy}{dx} = 6x - \frac{y}{x} - \frac{1}{x} \quad \checkmark$$

[3]

$$(b) \quad x + xy - 2x^3 = 2$$

$$\Rightarrow xy = 2 + 2x^3 - x \quad \checkmark$$

$$\Rightarrow y = \frac{2}{x} + 2x^2 - 1 \quad \checkmark$$

$$\Rightarrow y = 2x^{-1} + 2x^2 - 1$$

$$\frac{dy}{dx} = -2x^{-2} + 4x \quad \checkmark$$

[3]

$$(c) \quad \text{In (i) we got } \frac{dy}{dx} = 6x - \frac{y}{x} - \frac{1}{x} \quad \checkmark \text{ from fn}$$

$$y = \frac{2}{x} + 2x^2 - 1$$

$$\Rightarrow \frac{dy}{dx} = 6x - \frac{\left(\frac{2}{x} + 2x^2 - x\right)}{x} - \frac{1}{x} \quad \checkmark$$

[4]

$$= 6x - \left(\frac{2 + 2x^3 - x}{x^2}\right) - \frac{1}{x} \quad \checkmark$$

$$= 6x - (2x^{-2} + 2x - x^{-1}) - x^{-1} \quad \checkmark$$

$$= 6x - 2x^{-2} - 2x + x^{-1} - x^{-1} \quad \checkmark$$

$$= 4x - 2x^{-2} = \frac{dy}{dx} \text{ in (b).} \quad \checkmark$$

QUESTION 4

$$(a) \int \frac{x^2 - 2x}{\sqrt{x}} dx$$

$$= \int \left(\frac{x^2}{\sqrt{x}} - \frac{2x}{\sqrt{x}} \right) dx \checkmark$$

$$= \int (x^{3/2} - 2x^{1/2}) dx \checkmark$$

$$= \frac{x^{5/2}}{5/2} - 2 \frac{x^{3/2}}{3/2} + C \checkmark$$

$$[4] = \frac{2}{5} x^{5/2} - \frac{4}{3} x^{3/2} + C \checkmark$$

$$(b) \int_1^2 x e^x dx$$

$$u = x \quad dv = e^x dx$$

$$du = dx \quad v = e^x \checkmark$$

$$\therefore \int_1^2 x e^x dx = uv - \int v du \checkmark$$

$$= x e^x \Big|_1^2 - \int_1^2 e^x dx \checkmark$$

$$= (x e^x - e^x) \Big|_1^2 \checkmark$$

$$= (2e^2 - e^2) - (e^1 - e^1)$$

$$= e^2 \quad [6] \checkmark$$

QUESTION 5

$$(a) \int \cos x e^{\sin x} dx$$

$$\text{let } u = \sin x \quad du = \cos x dx \checkmark$$

$$\therefore \int \cos x e^{\sin x} dx = \int e^u du \checkmark$$

$$= e^u + C \checkmark$$

$$[4] = e^{\sin x} + C \checkmark$$

$$5(b) \quad y = x^2 - 2, \quad y = -x$$

Intersection:

$$x^2 - 2 = -x$$

$$x^2 + x - 2 = 0$$

$$(x+2)(x-1) = 0$$

$$x = -2, 1 \checkmark$$

$$\forall x \in (-2, 1) \quad y = -x > y = x^2 - 2 \checkmark$$

$$\therefore A = \int_a^b (f(x) - g(x)) dx \checkmark$$

where $f(x) > g(x) \quad \forall x \in (a, b)$

$$\therefore A = \int_{-2}^1 ((-x) - (x^2 - 2)) dx \checkmark$$

$$= \int_{-2}^1 (-x - x^2 + 2) dx \checkmark$$

$$= -\frac{x^2}{2} - \frac{x^3}{3} + 2x \Big|_{-2}^1 \checkmark$$

$$= \left(-\frac{1}{2} - \frac{1}{3} + 2\right) - \left(-2 + \frac{8}{3} - 4\right)$$

$$= \frac{9}{2} \checkmark$$

[6]

problem 6

$$(a) \int \frac{1}{x^2+x} dx = \int \frac{1}{x(x+1)} dx$$

$$\frac{1}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1} \quad \checkmark$$

$$\text{when } x=0, A = \frac{1}{1} = 1 \quad \checkmark$$

$$x=-1, B = \frac{1}{-1} = -1 \quad \checkmark$$

$$\therefore \frac{1}{x(x+1)} = \frac{1}{x} - \frac{1}{x+1} \quad \checkmark \quad [6]$$

integrating both side w.r.t x
gives

$$\begin{aligned} \int \frac{1}{x(x+1)} dx &= \ln|x| - \ln|x+1| + C \quad \checkmark \\ &= \ln\left|\frac{x}{x+1}\right| + C \quad \checkmark \end{aligned}$$

$$(b) F(x) = \int_2^{x^2} (\cos t) dt$$

$$\text{let } u = x^2 \quad \frac{du}{dx} = 2x \quad \checkmark$$

$$F'(x) = \frac{dF}{du} \cdot \frac{du}{dx} \quad \checkmark$$

$$= \frac{d}{du} \left(\int_2^u \cos t \, dt \right) \frac{du}{dx} \quad \checkmark$$

$$= (\cos u)(2x) \quad \checkmark$$

$$= (\cos x^2)(2x) \quad \checkmark$$

[4]